FROM CHEREDNIK ALGEBRAS TO KNOT HOMOLOGY VIA CUSPIDAL D-MODULES

XINCHUN MA

Abstract. We show that the triply-graded Khovanov-Rozansky homology of the torus knot $T_{m,n}$ can be recovered from the finite-dimensional representation $\mathrm{L}_{m/n}$ of the rational Cherednik algebra at slope m/n, endowed with the Hodge filtration coming from the cuspidal character D-module $\mathbf{N}_{m/n}$. Our approach involves expressing the associated graded of $\mathbf{N}_{m/n}$ in terms of a DG module closely related to the action of the elliptic Hall algebra on the equivariant K-theory of the Hilbert scheme of points on $\mathbb{C}^2$, thereby proving the rational master conjecture. As a corollary, we identify the Hodge filtration with the inductive and algebraic filtrations on $\mathrm{L}_{m/n}$.

Contents

1. Introduction 1
2. Cherednik algebras, mirabolic D-modules, Hilbert schemes 5
3. Hodge filtrations on cuspidal D-modules 12
4. K-theory of Hilbert schemes and bigraded characters 20
5. Proofs of the main results 30
Appendix A. Fourier transforms of cuspidal D-modules 36
Appendix B. Examples 37
References 40

1. Introduction

1.1. **Rational Cherednik algebras and link homology.** Recent years have seen extraordinary connections between seemingly unrelated mathematical objects across different fields, indexed by pairs of coprime natural numbers m, n. Topologically, there is the (m, n)-torus knot $T_{m,n}$, which winds m times around a circle in the interior of the torus, and n times around its axis of rotational symmetry. Surprisingly, the Khovanov-Rozansky(KhR) homology of $T_{m,n}$ is captured by the finite-dimensional irreducible representation of the rational Cherednik algebra.

We first briefly recall the definition of KhR homology. For any element γ in the braid group, one can associate a Rouquier complex built on Soergel bimodules labeled by simple twists. For each $i \geq 0$, take the i-th Hochschild homology of each term of the Rouquier complex to form a new complex. The sum over i of cohomology of these new complexes, usually denoted by $\mathrm{HHH}(\overline{\gamma})$, only depends on the link closure $\overline{\gamma}$ of the braid γ and defines the KhR homology of $\overline{\gamma}$ [Kho07]. Furthermore, under this definition, the KhR homology is naturally triply graded by the Hochschild homological grading, the internal grading on Soergel bimodules and the cohomological grading.

For $n > 0$, we let $\mathfrak{h}$ denote the $(n-1)$-dimensional standard representation of the symmetric group S_n and $\mathcal{D}(\mathfrak{h})$ the ring of differential operators on $\mathfrak{h}$. The rational Cherednik algebra (RCA) [EG02] $\mathrm{H}_c(S_n, \mathfrak{h})$, also known as the rational degeneration of the double affine Hecke algebra, is a deformation of $\mathcal{D}(\mathfrak{h}) \rtimes S_n$ at a complex parameter c. It turns out that only when $c = \frac{m}{n}$ for coprime m, n does $\mathrm{H}_c(S_n, \mathfrak{h})$ have finite-dimensional representations; in that case, there is a unique

irreducible one [BEG03], which we denote by $\mathrm{L}_{m/n}$. There is Euler grading on $\mathrm{L}_{m/n}$ induced by the action of the Euler field $\mathrm{h}_{m/n}$ (cf (17)). This grading induces a direct sum decomposition $\mathrm{L}_{m/n} = \bigoplus_{\ell\in\mathbb{Z}} \mathrm{L}_{m/n}(\ell)$.

Gorsky, Oblomkov, Rasmussen and Shende proposed the following:

Conjecture 1.1. *([GORS14, Conjecture 1.2]) For positive coprime integers m, n, there exists a filtration F on $\mathrm{L}_{m/n}$ compatible with the order filtration on $\mathrm{H}_{m/n}(S_n, \mathfrak{h})$ and the $\mathrm{h}_{m/n}$-grading such that there is an isomorphism of triply graded vector spaces:*

$$\bigoplus_{i,j,k} \mathrm{HHH}^{i,j,k}(T_{m,n}) \cong \bigoplus_{i} \mathrm{Hom}_{S_n}\left(\wedge^i\mathfrak{h}, \bigoplus_{j,k} \mathrm{gr}_j^F\left(\mathrm{L}_{m/n}(k)\right)\right). \tag{1}$$

The following gradings in the LHS and RHS, respectively, match:

Hochschild homological a-grading $\leftrightarrow$ degree of $\wedge^\bullet\mathfrak{h}$

internal q-grading $\leftrightarrow$ $\mathrm{h}_{m/n}$-grading by the weights of the $\mathrm{h}_{m/n}$-action

cohomological t-grading $\leftrightarrow$ filtration on $\mathrm{L}_{m/n}$

A bigraded isomorphism, disregarding the information of the t-grading, is established in [GORS14] by comparing explicit calculations of both sides of (1), as obtained in [Jon87] and [BEG03].

For every coprime pair (m, n), $m > n > 0$, we define a Hodge filtration on $\mathrm{L}_{m/n}$ and show that:

Theorem A. *(= Theorem 5.7) When $m > n$ such that $(m, n) = 1$, Conjecture 1.1 holds with respect to the Kazhdan filtration F^K associated to the Hodge filtration and $\mathrm{h}_{m/n}$.*

For the cases when $0 < m < n$, see Corollary 1.3.

1.2. **Hilbert schemes and the rational master conjecture.** Write $c = \frac{m}{n}$ for positive coprime integers m, n. Our approach is to construct a coherent sheaf $\mathcal{G}_c$ on $\mathrm{Hilb}^n := \mathrm{Hilb}^n(\mathbb{C}^2)$, the Hilbert scheme of n points on $\mathbb{C}^2$. The equivariant K-theory class of this sheaf computes the graded dimension of the right hand side of isomorphism (1). Specifically, consider the $\mathbb{C}^*\times\mathbb{C}^*$-action on Hilb^n induced by the natural action on $\mathbb{C}^2$. The $\mathbb{C}^*\times\mathbb{C}^*$-equivariant K-theory of Hilb^n, denoted by $K^{\mathbb{C}^*\times\mathbb{C}^*}(\mathrm{Hilb}^n)$, is a module over $K^{\mathbb{C}^*\times\mathbb{C}^*}(pt) \cong \mathbb{C}[q^\pm, t^\pm]$. The generalized McKay correspondence, proved by Haiman, [Hai01] implies an isomorphism

$$\bigoplus_n \mathbb{C}(q,t) \otimes_{\mathbb{C}[q^\pm,t^\pm]} K^{\mathbb{C}^*\times\mathbb{C}^*}(\mathrm{Hilb}^n) \cong \mathbb{C}(q,t)[z_1, z_2, \dots]^{S_\infty}. \tag{2}$$

The variables t, q correspond to the j, k gradings in (1) respectively. The i grading comes from the coefficient of the Schur polynomial labeled by the S_n-representation $\wedge^i\mathfrak{h}$.

More generally, Gorsky and Neguţ conjectured the following character formula in terms of the action of a generator $P_{m,n}$ of the elliptic Hall algebra (EHA) on its polynomial representation, identified with (2).

Conjecture 1.2. [GN15, Conjecture 5.5] *For positive coprime integers m, n, the bigraded Frobenius character of $\mathrm{L}_{m/n}$ with respect to an appropriate filtration and the $\mathrm{h}_{m/n}$-grading is given by*

$$\mathrm{ch}_{S_n\times\mathbb{C}^*\times\mathbb{C}^*}(\mathrm{L}_{m/n}) = (P_{m,n}\cdot 1)(qt, q^{-1}t). \tag{3}$$

Theorem B. *(= Theorem 5.5) When $m > n$ such that $(m, n) = 1$, Conjecture 1.2 holds with respect to the Kazhdan filtration F^K.*

For the cases when $0 < m < n$, also see Corollary 1.3. In the case of $m = n + 1$, $\mathrm{L}_{\frac{n+1}{n}}$ is isomorphic to the *diagonal harmonics* and the action of $P_{m,n}$ can be identified with that of the nabla operator [BG98]. Therefore, Conjecture 1.2 may be viewed as a rational generalization of

the master conjecture [GH96] proved by Haiman [Hai02] via the geometry of the Procesi bundle. We explicitly compute the left hand side of (3) by computing the bigraded Frobenius characters of fibers of the sheaf $\mathcal{G}_c$ at fixed points and using the localization formula. Comparing it with the computation by Neguţ [Neg15a] of the right hand side of (3) leads to our proof of Conjecture 1.2.

This result relates to the previous section in the following way. The computation of the Khovanov-Rozansky homology for torus links had long been a challenging open problem. It was finally addressed by Elias, Hogancamp and Mellit [Hog17, EH19, Hog18, Mel22] through recursive methods and concluded with a shuffle conjecture style formula. From this formula, it follows that the Euler characteristic of $\mathrm{HHH}(T_{m,n})$ equals the knot superpolynomial of $T_{m,n}$ defined in terms of $P_{m,n} \cdot 1$, [Che13, GN15]. Therefore, Theorem B implies Theorem A.

1.3. **DG flag commuting schemes.** To construct the coherent sheaf $\mathcal{G}_c$ on Hilb^n, we consider the cuspidal mirabolic D-module $\overline{\mathbf{N}}_c$ on $\mathfrak{sl}_n \times \mathbb{C}^n$, closely related to the cuspidal character sheaf in the sense of Lusztig [Lus86]. A result of Calaque, Enriques and Etingof [CEE09] implies that $(\mathrm{L}_c)^{S_n}$ is the *quantum Hamiltonian reduction* of $\overline{\mathbf{N}}_c$. We take the associated graded of $\overline{\mathbf{N}}_c$ with respect to the Hodge filtration (Hodge associated graded for short henceforth) within the framework of Saito's theory of mixed Hodge modules [Sai90]. The descent of $\mathrm{gr}(\overline{\mathbf{N}}_c)$ is the desired coherent sheaf on Hilb^n.

As noted by Neguţ in [Neg15a], the action of the generator $P_{m,n}$ on $K^{\mathbb{C}^* \times \mathbb{C}^*}(\mathrm{Hilb}^n(\mathbb{C}^2))$ is not visible using the classical Nakajima correspondence, and the notion of a flag Hilbert scheme turns out to be necessary. However, it is well-known that the naive flag Hilbert scheme is highly singular [Che98]. A remedy is to rather consider the DG flag Hilbert scheme $\mathrm{FHilb}^n_{\mathrm{dg}}(\mathbb{C}^2)$, defined independently in [Gin12] and [Neg15a].

In [Gin12], Ginzburg studies the isospectral commuting variety by expressing the Hodge associated graded of the Harish-Chandra D-module in terms of the DG flag commuting variety. In type A, this facilitates the study of the Procesi bundle from a different perspective [Gor12, BG13].

We follow the method of [Gin12] and naturally obtain the *cuspidal DG module* $\mathcal{A}_c$ in the computation of $\mathrm{gr}(\overline{\mathbf{N}}_c)$. Motivated by the similar origins of $\mathcal{A}_0$ and the DG flag commuting variety, we expect that $\mathcal{A}_0$, whose definition is valid for all types, might give the correct definition for the DG nilpotent flag commuting scheme.

We also define a Catalan DG module. Adapting the method of [BG13], we compute the equivariant K-theory classes corresponding to the pushforward of these two DG modules. The one corresponding to the Catalan DG module encodes the information of the q,t-Catalan number. By the shuffle algebra machinary developed by Neguţ, we identify these two classes and expect that the sheaves themselves are isomorphic (Conjecture 4.6). While this paper was under editing, Gorsky and Neguţ generously shared a copy of their preprint [GN24]. Their Conjecture 2.2 is a non-derived version of our Conjecture 4.6.

1.4. **Filtrations and future directions.** Regarding Conjecture 1.1, the authors of [GORS14] have proposed several filtrations on $\mathrm{L}_{m/n}$:

(a) The inductive filtration F^{ind} is defined inductively using the shift functor from $\mathrm{H}_{m/n}$-modules to $\mathrm{H}_{\frac{m}{n}+1}$-modules and the "flipping" isomorphism $(\mathrm{L}_{m/n})^{S_n} \cong (\mathrm{L}_{\frac{n}{m}})^{S_m}$.
(b) The algebraic filtration F^{alg} is defined using powers of Chern classes in the equivariant cohomology of the compactified Jacobian of the planar singular curve $y^m = x^n$.
(c) The geometric filtration F^{geom} is defined in terms of the perverse filtration and the cohomological grading on the cohomology ring as in (b).

The inductive filtration and the Hodge filtrations are defined on L_c for $m > n$ and on $\mathrm{L}_c^{S_n}$ for all $m > 0$ while the algebraic filtration is well-defined on L_c for all $m > 0$.

The equality $F^{\mathrm{alg}} = F^{\mathrm{geom}}$ is proved in [OY16] and the equality $F^{\mathrm{ind}} = F^{\mathrm{alg}}$ is proved in [Ma24].

Theorem C. *(= Proposition 5.9)*

$$\begin{cases} F_i^K \mathrm{L}_{m/n} = F_i^{\mathrm{ind}} \mathrm{L}_{m/n} & \forall i \in \mathbb{Z} \text{ when } m > n \\ F_i^K (\mathrm{L}_{\frac{m}{n}})^{S_n} = F_i^{\mathrm{ind}} (\mathrm{L}_{\frac{m}{n}})^{S_n} & \forall i \in \mathbb{Z} \end{cases}$$

As a corollary of Theorem C, we show that:

Corollary 1.3. *(= Proposition 5.10) For all integers $m > 0$ coprime to n, with respect to the algebraic filtration, Conjectures 1.1 and 1.2 hold.*

There are two natural generalizations of our setting: allowing m, n to be not necessarily coprime or replacing $\mathrm{Hilb}^n(\mathbb{C}^2)$ by Gieseker varieties. The former is related to the study of RCA representations of minimal support and torus link homology ([EGL15]). The latter is related to the study of representations of a quantized Gieseker moduli algebra ([EKLS21]) and the study of higher Catalan numbers ([GSV23]). Despite the current absense of definitions of the inductive, algebraic and geometric filtrations in these settings, we believe that a notion of Hodge filtration is still available, and via a similar method, bigraded Frobenius characters can be computed and related to the corresponding link invariants or Catalan statistics.

On the other hand, it is conjectured that the stable envelopes on $\mathrm{Hilb}^n(\mathbb{C}^2)$ are closely related to Verma modules of Cherednik algebras [GN17, Conjecture 6.5]. In general, the connection between Verma modules over quantized symplectic resolutions and categorical stable envelopes is studied by Bezrukavnikov and Okounkov [BO] using positive characteristic technique. We believe the Hodge filtration might be the correct candidate; however, it is neither clear nor has it been explored which explicit D-modules correspond to the Verma modules. We plan to investigate these directions in future work.

1.5. **Structure of the paper.** Section 2 covers preliminaries. We introduce the category $\mathcal{O}$ for RCA, mirabolic $D(\mathfrak{sl}_n \times \mathbb{C}^n)$-modules and coherent sheaves on Hilb^n. We define the quantum Hamiltonian reduction and the Gordon-Stafford functor connecting these objects.

In Section 3, we study Hodge filtrations on cuspidal mirabolic D-modules. We first restrict $\overline{\mathbf{N}}_c$ to a Borel subalgebra. The resulting Hodge module is explicitly extended from an integrable connection across a simple normal crossing divisor. We extend this structure back to $\overline{\mathbf{N}}_c$ via parabolic induction and D-module pushforward along the Springer resolution. We express $\mathrm{gr}(\overline{\mathbf{N}}_c)$ as a pushforward of the cuspidal DG module $\mathcal{A}_c$.

In Section **??**, we compute the equivariant K-theory class on Hilb^n corresponding to $\mathrm{gr}(\overline{\mathbf{N}}_c)$.

In Section **??**, we introduce the Catalan DG module $\mathcal{A}'_c$, analogous to $\mathcal{A}_c$. We show the equivariant K-theory classes on Hilb^n corresponding to $\mathcal{A}_c$ and $\mathcal{A}'_c$ are both equal to $P_{m,n} \cdot 1$.

In Section 5, we conclude the proofs of the main theorems.

Appendix A is a digression on the invariance of $\overline{\mathbf{N}}_c$ under the Fourier transform.

Appendix B compiles examples of K-theory computations for $\mathcal{A}_c$ and $\mathcal{A}'_c$.

1.6. **Acknowledgements.** I would like to express my deepest gratitude to my advisor, Victor Ginzburg, for suggesting this problem. His previous work forms the foundation of almost all the arguments presented here, and this paper would not have been possible without his guidance and inspiration. I am very grateful to Andrei Neguţ, especially for pointing out the reference [Neg22, (2.34) and (2.35)], which plays an essential role in the proofs in Section 4.8. I thank Eugene Gorsky for valuable discussions and for giving me the opportunities to present this work while it was still in progress. I thank Oscar Kivinen and Minh-Tâm Trinh for helpful discussions at an early stage of this project and Kostya Tolmachov for providing references on $\mathbb{C}$-Hodge modules.

2. Cherednik algebras, mirabolic D-modules, Hilbert schemes

Fix a positive integer n. We work over $\mathbb{C}$ throughout, and set up the following notations:
$\overline{G} = \mathrm{GL}_n$ with Lie algebra $\overline{\mathfrak{g}} := \mathfrak{gl}_n$;
$G = \mathrm{SL}_n$ with Lie algebra $\mathfrak{g} = \mathfrak{sl}_n$;
$\overline{T}$: a maximal torus in $\overline{G}$ with Lie algebra $\overline{\mathfrak{h}} \subset \overline{\mathfrak{g}}$, consisting of diagonal matrices;
$T = \overline{T} \cap G$ and $\mathfrak{h} = \overline{\mathfrak{h}} \cap \mathfrak{g}$;
$W = S_n$: the Weyl group;
$S \subset W$: the set of reflections.

For any smooth algebraic variety X, we denote the sheaf of differential operators on X by $\mathcal{D}_X$ and write $\mathcal{D}(X) := \Gamma(X, \mathcal{D}_X)$. A filtration on a coherent $\mathcal{D}_X$-module M is called good if the associated sheaf $\widetilde{\mathrm{gr}}(M)$ on T^*X is coherent. The support of $\widetilde{\mathrm{gr}}(M)$ in T^*X is independent of the choice of a good filtration, which is called the singular support of M and denoted by $SS(M)$.

2.1. **Rational Cherednik algebras.** For any $c \in \mathbb{C}$, the rational Cherednik algebra $\mathrm{H}_c := \mathrm{H}_c(W, \mathfrak{h})$ is defined as the $\mathbb{C}$-algebra generated by $\mathfrak{h}$, $\mathfrak{h}^*$ and W with relations

$$[x, x'] = [y, y'] = 0, \quad wxw^{-1} = w(x), \quad wyw^{-1} = w(y)$$
$$[y, x] = x(y) - \sum_{s \in S} c\langle \alpha_s, y\rangle \langle \alpha_s^\vee, x\rangle s$$

where $x, x' \in \mathfrak{h}^*$, $y, y' \in \mathfrak{h}$, $w \in W$. Here α_s, resp. $\alpha_s^\vee$, is the root, resp. coroot, associated to s and $\langle -, - \rangle$ is the pairing between $\mathfrak{h}$ and $\mathfrak{h}^*$.

For an irreducible W-representation μ, we regard it as a $S(\mathfrak{h}) \rtimes W$-representation by letting the augmented ideal $S(\mathfrak{h})_+$ act trivially and define the Verma module

$$M_c(\mu) = \mathrm{H}_c \otimes_{S(\mathfrak{h}) \rtimes W} \mu.$$

The Verma module $M_c(\mu)$ has a unique nonzero irreducible quotient $\mathrm{L}_c(\mu)$.

Example 2.1. When $c = 0$, $\mathrm{H}_0 = \mathcal{D}(\mathfrak{h}) \rtimes W$ whose polynomial representation $\mathbb{C}[\mathfrak{h}] \cong M_0(\mathrm{triv})$ is irreducible.

Theorem 2.1. *([BEG03, Theorem 1.2]) When $c = \frac{m}{n}$ for positive integer m coprime to n, the unique irreducible finite-dimensional representation of H_c is $\mathrm{L}_c := \mathrm{L}_c(\mathrm{triv})$. Moreover, H_c has finite-dimensional representations only when $c = \frac{m}{n}$ for integer m coprime to n.*

Let $\mathcal{O}(\mathrm{H}_c)$ denote category $\mathcal{O}$ for RCA, which is a full subcategory of H_c-mod whose objects are finitely generated over H_c and such that the $S(\mathfrak{h})$ action is locally nilpotent [GGOR03]. As μ runs over irreducible W-representations, the modules $\mathrm{L}_c(\mu)$ give a complete list of irreducible objects in $\mathcal{O}(\mathrm{H}_c)$.

Let $\mathrm{e} := \frac{1}{n!}\sum_{w \in W} w$ be the symmetrizing idempotent in H_c. Inside H_c, we have the isomorphisms $\mathrm{e}\mathbb{C}[\mathfrak{h}] \cong \mathbb{C}[\mathfrak{h}]^W$ and $\mathrm{e}S(\mathfrak{h}) \cong S(\mathfrak{h})^W$. The spherical Cherednik algebra is defined by $\mathrm{A}_c := \mathrm{e}\mathrm{H}_c\mathrm{e}$, specializing to $\mathrm{A}_0 = \mathcal{D}(\mathfrak{h})^W$. By [BE09, Corollary 4.2], for all c satisfying

$$c \notin \{\frac{a}{b} \in (-1, 0) | a, b \in \mathbb{Z}, 2 \le b \le n\} \tag{4}$$

there is a Morita equivalence

$$\mathrm{H}_c-\mathrm{mod} \to \mathrm{A}_c-\mathrm{mod}, \quad M \mapsto \mathrm{e}M.$$

Let $\mathcal{O}(\mathrm{A}_c)$ denote the category $\mathcal{O}$ of the spherical Cherednik algebra.

Let R^+ be the set of positive roots. Write $\delta = \prod_{\alpha \in R^+} \alpha$. Let $\mathfrak{h}_r = \{\delta \neq 0\}$ denote the regular locus of $\mathfrak{h}$ of diagonal matrices with pairwise distinct entries. The action of H_c on $\mathbb{C}[\mathfrak{h}] \cong \mathrm{H}_c \otimes_{\mathbb{C}[\mathfrak{h}^*] \rtimes W} \mathbb{C}$ gives the Dunkl embedding of H_c into $\mathcal{D}(\mathfrak{h}_r) \rtimes W$. In particular, assume $\{x_i\}$ is a basis of $\mathfrak{h}^*$ and

$\{y_i\}$ is its dual basis of $\mathfrak{h}$. Let (ij) denote the reflection swapping i and j. The Dunkl operator associated to y_i is

$$D_{y_i,c} := \partial_{x_i} - c\sum_{s\in S}\frac{\alpha_s(x_i)}{\alpha_s}(1-s) = \partial_{x_i} - c\sum_{j\neq i}\frac{1-(ij)}{x_i-x_j}.$$

Localized at δ, we have that $(\mathrm{H}_c)_\delta \cong \mathcal{D}(\mathfrak{h}_r)\rtimes W$. Consider the algebra automorphism on $(\mathrm{H}_c)_\delta$ defined by sending

$$x\mapsto x,\quad D_{y_i,c}\mapsto D_{y_i,-c},\quad w\mapsto \mathrm{sign}(w)w$$

which restricts to an isomorphism $\mathrm{A}_c \to \delta^{-1}\mathrm{A}_{-c-1}\delta$ ([GGS09, 5.6, 5.8]). This induces the following equivalence of categories, which we will use in Section 2.5:

$$\Omega_c = \Omega_{-c-1}^{-1} : \mathrm{A}_d\text{−mod} \cong \mathrm{A}_{-c-1}\text{−mod}.$$

2.2. **Quantum Hamiltonian reduction.** Let $V=\mathbb{C}^n$ on which $\overline{G}$ acts by left multiplication and put $\mathfrak{G}=\mathfrak{g}\times V$. Let $\tau:\overline{\mathfrak{g}}\to\mathcal{D}(\mathfrak{G})$ denote the embedding induced by differentiating the diagonal action of $\overline{G}$ on $\mathfrak{G}$. For any $f\in\mathcal{O}(\mathfrak{G})$, an element $g\in\overline{G}$ acts by

$$(g\cdot f)(x) = f(g^{-1}\cdot x). \tag{5}$$

In particular, the identity matrix $\mathbf{1}\in\overline{\mathfrak{g}}$ acts as the operator $\tau(\mathbf{1}) = -\sum_{i=1}^n v_i\partial_{v_i}$ where v_i's are the standard coordinates on $\mathbb{C}^n$.

For a constant $d\in\mathbb{C}$, define the shifted embedding τ_d by

$$\tau_d : \overline{\mathfrak{g}}\to\mathcal{D}(\mathfrak{G}),\quad x\to\tau(x)-d\cdot\mathrm{tr}(x). \tag{6}$$

Later, e.g, in Section 2.3.3, we will take d to be $-c$.

Identify $\mathfrak{g}\cong\mathfrak{g}^*$ via the Killing form. Let $\mathscr{N}\subset\mathfrak{g}$ denote the nilpotent cone. The following subvariety of $T^*\mathfrak{G}\cong\mathfrak{g}\times\mathfrak{g}\times V\times V^*$is known to be Lagrangian:

$$\Lambda := \{(X,Y,i,j)\in\mathfrak{g}\times\mathscr{N}\times V\times V^* | [X,Y]+ij=0\}.$$

Definition 2.1. ([FG10, GG06]) A finitely generated $\mathcal{D}_{\mathfrak{G}}$-module M of regular singularities is called **mirabolic** if $SS(M)\subset\Lambda$.

Let $\mathfrak{C}(\mathfrak{G})$ denote the category of mirabolic D-modules.

Fix a nonzero element $\mathrm{vol}\in\bigwedge^n V^*$. Following [BFG06, (5.3.2)], consider the regular function on $\mathfrak{G}$ defined by:

$$s(x,v) = \langle \mathrm{vol}, v\wedge xv\wedge\cdots\wedge x^{n-1}v\rangle. \tag{7}$$

Take $\mathfrak{G}_{\mathrm{cyc}} := \{(x,v)\in\mathfrak{G} | s(x,v)\neq 0\}$, the locus where v is a cyclic vector for x, i.e. $\mathbb{C}[x]v=V$. By [BFG06, Lemma 5.3.3], the composition of the projection $\mathfrak{G}_{\mathrm{cyc}}\to\mathfrak{g}$ and the Chevalley map $\mathfrak{g}\to\mathfrak{g}//G\cong\mathfrak{h}/W$ is a principal $\overline{G}$-bundle and the diagonal $\overline{G}$-action on $\mathfrak{G}_{\mathrm{cyc}}$ is free along the fibers. Therefore we have an algebra isomorphism

$$\iota : \mathbb{C}[\mathfrak{G}_{\mathrm{cyc}}]^{\overline{G}}\cong\mathbb{C}[\mathfrak{h}]^W. \tag{8}$$

Note that $s^{-d}\in\mathbb{C}[\mathfrak{G}_{\mathrm{cyc}}]^{\tau_d(\overline{\mathfrak{g}})}$. In fact, $\mathbb{C}[\mathfrak{G}_{\mathrm{cyc}}]^{\tau_d(\overline{\mathfrak{g}})} = \mathbb{C}[\mathfrak{G}_{\mathrm{cyc}}]^G s^{-d}$.

By [BG15, Theorem 1.2.4], any mirabolic D-module is locally finite as a $\tau(\overline{\mathfrak{g}})$-module. In particular, under the $\tau(\mathbf{1})$ action, any mirabolic D-module $\mathbf{M}$ decomposes into a direct sum

$$\Gamma(\mathbf{M}) = \bigoplus_{d+\mathbb{Z}\in\mathbb{C}/\mathbb{Z}}\Gamma(\mathbf{M})(d+\mathbb{Z})$$

where

$$\mathbf{M}(d+\mathbb{Z}) = \{v\in\Gamma(\mathbf{M}), (\tau(\mathbf{1})-d-k)^i v=0, \text{for some } k\in\mathbb{Z} \text{ and } i\gg 0\}.$$

Given $d\in\mathbb{C}$, let $\mathscr{C}_d(\mathfrak{G})$ be the full subcategory of $\mathscr{C}(\mathfrak{G})$ whose objects satisfy $\Gamma(\mathbf{M})=\Gamma(\mathbf{M})(d+\mathbb{Z})$.

Theorem 2.2. *([GG06, Theorem 1.3.1], [GGS09, Theorem 8.1]) The radial part map*

$$\mathcal{D}(\mathfrak{G})^{\overline{G}} \to \mathcal{D}(\mathfrak{h}/W), \quad u \mapsto \left[\mathbb{C}[\mathfrak{h}/W] \ni f \mapsto \iota\left(s^d u\left(s^{-d}\iota^{-1}(f)\right)\right)\right]$$

induces an isomorphism

$$\mathscr{H}_d : (\mathcal{D}(\mathfrak{G})/\mathcal{D}(\mathfrak{G})\tau_d(\overline{\mathfrak{g}}))^{\overline{G}} \cong \mathrm{A}_{d-1}. \tag{9}$$

Thus, there is the quantum Hamiltonian reduction functor

$$\mathbb{H}_d : \mathscr{C}_d(\mathfrak{G}) \to \mathcal{O}(\mathrm{A}_{d-1}), \quad \mathbf{M} \mapsto \Gamma(\mathfrak{G}, \mathbf{M})^{\tau_d(\overline{\mathfrak{g}})}.$$

Moreover, the kernel forms a Serre subcategory and

$$\mathscr{C}_d(\mathfrak{G})/\mathrm{Ker}(\mathbb{H}_d) \cong \mathcal{O}(\mathrm{A}_{d-1}).$$

Remark 2.2. The shift by "-1" in the subscript of A_{d-1} is consistent with the classical result of Harish-Chandra that

$$\mathcal{D}(\mathfrak{g})^{G} \to \mathcal{D}(\mathfrak{h}/W), \quad u \mapsto \left[\mathbb{C}[\mathfrak{h}/W] \ni f \mapsto \delta(u|_{\mathbb{C}[\mathfrak{h}/W]})\left(\delta^{-1}f\right)\right]$$

induces an isomorphism $(\mathcal{D}(\mathfrak{g})/\mathcal{D}(\mathfrak{g})\mathrm{ad}(\mathfrak{g}))^{G} \cong D(\mathfrak{h})^{W}$. Here δ is the product of all positive roots and the restriction $u|_{\mathbb{C}[\mathfrak{h}/W]}$ is taken with respect to the Chevalley isomorphism $C[\mathfrak{g}]^G \cong \mathbb{C}[\mathfrak{h}]^W$.

2.3. Cuspidal mirabolic D-modules.

2.3.1. *Functors on D-modules.* Let $f : X \to Y$ be a morphism of smooth algebraic varieties. We write $f_\dagger, f_!, f^\dagger$ to denote the derived pushforward, proper pushforward and pullback functors of holonomic D-modules respectively, cf. [HTT08, Section 1.5].

When f is a locally closed embedding, one defines the minimal extension functor $f_{!*}$ to be the image under the canonical morphism $f_! \to f_\dagger$. For a locally closed subset $U \xrightarrow{i} X$, an integrable connection on U (i.e. a $\mathcal{O}_U$-coherent $\mathcal{D}_U$-module) is called clean if its minimal extension coincides with the extensions using $f_\dagger$ or $f_!$.

2.3.2. *Cuspidal integrable connections.* The zero fiber of the fibration $\mathfrak{G}_{\mathrm{cyc}} \to \mathfrak{h}/W$ is

$$U := \{(x, v) \in \mathscr{N} \times V | s(x, v) \neq 0\}.$$

The diagonal $\overline{G}$ action on U is transitive and free. Hence $\pi_1(U) = \mathbb{Z}$. We have that $g \cdot s = \det(g)^{-1}s$ for any $g \in \overline{G}$ and thus every simple local system on U of finite-order monodromy is spanned by the horizontal section s^a for some rational number a. We will denote by $\mathcal{E}_a$ the integrable connection corresponding to such a local system under the Riemann-Hilbert correspondence.

Let $\mathscr{N}_r \subset \mathscr{N}$ denote the regular nilpotent orbit. Then $\pi_1(\mathscr{N}_r) = \mathbb{Z}/n\mathbb{Z}$ and every simple local system on $\mathscr{N}_r$ of finite order monodromy corresponds to the representation of $\mathbb{Z}/n\mathbb{Z}$ defined by $e^{2\pi i b}$ for some $b \in \frac{1}{n}\mathbb{Z}$. Let $\mathcal{F}_b$ denote the integrable connection on $\mathscr{N}_r$ corresponding to such a local system. Both $\mathcal{E}_a$ and $\mathcal{F}_b$ are G-equivariant.

The projection $U \to \mathscr{N}$ has image inside $\mathscr{N}_r$ and the fibers of this projection are isomorphic to $\mathfrak{F} := \mathbb{C}^{n-1} \times \mathbb{C}^*$. The fibration

$$\begin{array}{ccc} \mathfrak{F} & \longrightarrow & U \\ & & \downarrow \\ & & \mathscr{N}_r \end{array}$$

induces an exact sequence

$$1 \to (\mathbb{Z} = \pi_1(\mathfrak{F})) \xrightarrow{n} (\mathbb{Z} = \pi_1(U)) \to (\pi_1(\mathscr{N}_r) = \mathbb{Z}/n\mathbb{Z}) \to 1.$$

It follows that $\mathcal{E}_a$ is the pullback of $\mathcal{F}_a$ on $\mathscr{N}_r$ for $a \in \frac{1}{n}\mathbb{Z}$.

2.3.3. *Cuspidal character/mirabolic D-modules.* Let $c = \frac{m}{n}$ for positive integer m coprime to n.

Definition 2.2.
- The cuspidal character $\mathcal{D}_{\mathfrak{g}}$-module of parameter c is the minimal extension of $\mathcal{F}_c$ to $\mathfrak{g}$, which we denote by $\mathbf{N}_c$.
- The cuspidal mirabolic $\mathcal{D}_{\mathfrak{G}}$-module of parameter c is the minimal extension of $\mathcal{E}_c$ to $\mathfrak{G}$, which we denote by $\overline{\mathbf{N}}_c$.

By [Lus86], $\mathcal{F}_c$ is clean. We also have the following:

Lemma 2.3. *As G-equivariant $\mathcal{D}_{\mathfrak{G}}$-modules, $\overline{\mathbf{N}}_c \cong \mathbf{N}_c \boxtimes \mathcal{O}_V$. Moreover*

$$SS(\overline{\mathbf{N}}_c) = \{(x,y) \in \mathscr{N} \times \mathscr{N} | [x,y] = 0\}_{\text{red}} \times V \tag{10}$$

where the subscript $_{\text{red}}$ *refers to taking the reduced structure.*

Proof. The function s has degree n along the direction of V. Since $c \cdot n = m$, the local system $\mathcal{E}_c$, defined by the horizontal section s^c, has no monodromy along the V direction. Therefore the minimal extension of $\mathcal{E}_c$ to $\mathscr{N}_r \times V$ is $\mathcal{F}_c \boxtimes \mathcal{O}_V$ and the first statement follows from the cleanness of $\mathcal{F}_c$. On the other hand, the singular support of a cuspidal character D-module is known to equal $\{(x,y) \in \mathscr{N} \times \mathscr{N} | [x,y] = 0\}_{\text{red}}$ ([**?**, Theorem 4.7]) and hence the second statement of the lemma follows. □

Theorem 2.4. *([CEE09, Theorem 9.19])*
- *There is an A_{-c-1}-action on $(\Gamma(\mathfrak{g}, \mathbf{N}_c) \otimes \mathrm{Sym}^m V)^G$.*
- *Under this action, $(\Gamma(\mathfrak{g}, \mathbf{N}_c) \otimes \mathrm{Sym}^m V)^G$ is isomorphic to $\Omega_c(\mathrm{eL}_c)$ (where Ω is defined in Section 2.1).*

Corollary 2.5. *As A_{-c-1}-modules, $\mathbb{H}_{-c}(\overline{\mathbf{N}}_c) \cong \Omega_c(\mathrm{eL}_c)$ (where H_{-c} is the quantum Hamiltonian reduction functor in Theorem 2.2).*

Proof. Follows from Lemma 2.3, Theorem 2.4 and the fact that $\mathbb{C}[V]^{\tau_{-c}(\mathbf{1})} = \mathrm{Sym}^m V$. □

2.4. **Hilbert schemes of points.** Let $\overline{\mathrm{Hilb}}^n(\mathbb{C}^2)$ denote the moduli space of ideals of colength n in $\mathbb{C}[x,y]$. It is a smooth and quasi-projective variety of dimension $2n$.

The Hilbert-Chow map $\overline{\mathrm{Hilb}}^n \to (\mathbb{C}^2)^n/S_n$, defined by sending a colength n ideal I to the subvariety defined by the quotient $\mathbb{C}[x,y]/I$, is a resolution of singularity. We let $\mathrm{Hilb}^n \subset \overline{\mathrm{Hilb}}^n(\mathbb{C}^2)$ denote the preimage of $\{(z_1, \cdots, z_n) \in (\mathbb{C}^2)^n, \sum_i z_i = (0,0)\}$ under the Hilbert-Chow map.

Define

$$\widetilde{\mathrm{Hilb}^n} := \{(X,Y,v) \in \mathfrak{g} \times \mathfrak{g} \times V | [X,Y] = 0, \mathbb{C}[X,Y]v = V\}_{\text{red}}.$$

The diagonal G-action on $\widetilde{\mathrm{Hilb}^n}$ is free and the resulting GIT quotient is $\widetilde{\mathrm{Hilb}^n}/\!\!/G = \mathrm{Hilb}^n$(see [Nak99]).

Let $\widetilde{\mathrm{Hilb}}_0^n = (\mathscr{N} \times \mathscr{N} \times V) \cap \widetilde{\mathrm{Hilb}^n}$, which is an open subvariety of the singular support (10). The GIT quotient $\mathrm{Hilb}_0^n := \widetilde{\mathrm{Hilb}}_0^n/\!\!/G$ is the zero fiber of the Hilbert-Chow map, known as the punctual Hibert scheme.

2.5. **The Gordon-Stafford functor.** Consider the order filtration on H_c defined by $\deg y = 1$ and $\deg x = \deg w = 0$. We will use this filtration on all subspaces of H_c henceforth. With respect to this filtration, we have that $\mathrm{gr}(\mathrm{A}_c) = \mathbb{C}[\mathfrak{h} \times \mathfrak{h}^*]^W$. We are interested in A_c-modules with good filtrations, in the sense that the associated graded is finitely generated as a $\mathbb{C}[\mathfrak{h} \times \mathfrak{h}^*]^W$ module.

In [GS05], Gordon and Stafford define a functor from the category of A_c-modules equipped with good filtrations to $\mathrm{Coh}^{\mathbb{C}^*\times\mathbb{C}^*}(\mathrm{Hilb}^n(\mathbb{C}^2))$, which completes the following diagram:

$$\begin{array}{ccc} ? & \overset{\mathrm{gr}}{\rightsquigarrow} & \mathcal{O}_{\mathrm{Hilb}^n} \\ {\scriptstyle GS}\uparrow & & \uparrow{\scriptstyle \text{Hilbert-Chow}} \\ \mathrm{A}_c & \overset{\mathrm{gr}}{\rightsquigarrow} & \mathbb{C}[\mathfrak{h}\times\mathfrak{h}^*]^{S_n} \end{array}$$

The definition of their functor GS is based on the Proj construction of the Hilbert scheme due to Haiman [Hai98, Proposition 2.6], which we recall as follows. Let $R_0 = \mathbb{C}[\mathfrak{h}]^{S_n}$ and $R_1 = \mathbb{C}[\mathfrak{h}]^{\mathrm{sgn}}$. The supscript ^{sgn} indicates taking the sign-isotypic component under the S_n action. For each $k \geq 1$, let $R_k = (R_1)^k$ be the product of k copies of R_1 in $\mathbb{C}[\mathfrak{h}]$. Define a graded ring $R = \oplus_{k\geq 0} R_k$. Then $\mathrm{Hilb}^n \cong \mathrm{Proj}(R)$. Therefore, any finitely generated graded R-module defines a coherent sheaf on Hilb^n.

Let $\mathrm{e}_- := \frac{1}{n!}\sum_{w\in W}\mathrm{sgn}(w)w$ be the skew-symmetrizing idempotent in H_c and $\delta = \prod_{\alpha\in R^+}\alpha$. We define two subspaces in $(\mathrm{H}_d)_\delta$:

$$ {}_{d+1}\mathrm{P}_d := \mathrm{e}\mathrm{H}_d\delta\mathrm{e}_-, \quad {}_d\mathrm{Q}_{d+1} := \mathrm{e}_-\delta^{-1}\mathrm{H}_{d+1}\mathrm{e}.$$

Both ${}_{d+1}\mathrm{P}_d$ and ${}_d\mathrm{Q}_{d+1}$ inherit from $(\mathrm{H}_d)_\delta$ the order filtration. For any $d \in \mathbb{C}$, the isomorphism [BEG03, Proposition 4.6]

$$\mathrm{e}\mathrm{H}_d\mathrm{e} \cong \mathrm{e}\delta^{-1}\mathrm{H}_{d+1}\delta\mathrm{e}$$

gives a $(\mathrm{A}_{d+1}, \mathrm{A}_d)$-module structure on ${}_{d+1}\mathrm{P}_d$ and a $(\mathrm{A}_d, \mathrm{A}_{d+1})$-module structure on ${}_d\mathrm{Q}_{d+1}$. Thus we can inductively define for any $k \in \mathbb{Z}_{>0}$,

$$ {}_{d+k}\mathrm{P}_d := {}_{d+k}\mathrm{P}_{d+k-1} \otimes_{\mathrm{A}_{d+k-1}} {}_{d+k-1}\mathrm{P}_d,$$

and similarly for ${}_{d-k}\mathrm{Q}_d$.

For two filtered modules M, N over a filtered ring R, the tensor product filtration on $M \otimes_R N$ is defined by

$$F_k(M\otimes_R N) = \mathrm{Im}\left(\sum_{i+j=k} F_iM \otimes F_jN \to M\otimes_R N\right).$$

We equip ${}_{d+k}\mathrm{P}_d$ and ${}_{d-k}\mathrm{Q}_d$ with tensor product filtrations.

For any A_{-d-1}-module L, one has that [GGS09, Proposition 5.8]

$$ {}_{d+k}\mathrm{P}_d \otimes_{\mathrm{A}_d} \Omega_{-d-1}L \cong \Omega_{-(d+k)-1}({}_{-(d+k)-1}\mathrm{Q}_{-d-1} \otimes_{\mathrm{A}_{-d-1}} L). \tag{11}$$

Moreover, ${}_{c+k}\mathrm{P}_c$ is a $(\mathrm{A}_{c+k}, \mathrm{A}_c)$-module and hence defines a shift functor

$$S_{c,k} : A_c\text{−mod} \to A_{c+k}\text{−mod}, \quad M \mapsto {}_{c+k}\mathrm{P}_c \otimes_{\mathrm{A}_c} M.$$

For a filtered module M, one equips $S_{c,k}(M)$ with the associated tensor product filtration.

Definition 2.3. Suppose M is an A_c-module equipped with a good filtration $F_\bullet$. The Gordon-Stafford functor associates to $(M, F_\bullet)$ a coherent sheaf on Hilb^n defined by

$$GS(M, F_\bullet) = \mathrm{Proj}\left(\mathrm{gr}\bigoplus_{k\geq 0} S_{c,k}M\right).$$

2.6. **Relating the functors.** The parameter c in this subsection can be any complex number.

2.6.1. *Compatibility of filtrations.* Let $\mathbf{M} \in \mathscr{C}(\mathfrak{G})$ be a mirabolic D-module with a good filtration F.

For any $\overline{G}$-module E, write

$$E^{\det^{-k}} := \{f \in E | g \cdot f = \det(g)^{-k} f, \forall g \in \overline{G}\}.$$

Define $\mathcal{D}_d(\mathfrak{G}) := \mathcal{D}(\mathfrak{G})/\mathcal{D}_d(\mathfrak{G})\tau_d(\overline{\mathfrak{g}})$. There is a homomorphism

$$\phi_M^k : \mathcal{D}_{-c}(\mathfrak{G})^{\det^{-k}} \otimes_{\mathrm{A}_{-c-1}} M^{\tau_{-c}(\overline{\mathfrak{g}})} \to M^{\tau_{-c-k}(\overline{\mathfrak{g}})} \tag{12}$$

given by the left multiplication of $\mathcal{D}(\mathfrak{G})$ on M. By [BG15, Theorem 1.3.5], ϕ_M^k is an isomorphism of A_{-c-k}-modules. In general, ϕ_M^k is not necessarily a filtered isomorphism.

Lemma 2.6. *ϕ_M^k is a filtered isomorphism if and only if*

$$\mathrm{gr}\,(S_{c,k}L) \cong \mathrm{gr}\left(\Omega_{-c-k-1} M^{\tau_{-c-k}(\overline{\mathfrak{g}})}\right). \tag{13}$$

Proof. Moreover, by [GGS09, Theorem 5.3(1)], when each of the rational numbers $-c-1, \cdots, -c-k-1$ satisfies the condition (4), there is an isomorphism

$$\mathcal{D}_{-c}(\mathfrak{G})^{\det^{-k}} \cong {}_{-c-k-1}Q_{-c-1} \tag{14}$$

of filtered modules with respect to the order filtrations inherited from $\mathcal{D}(\mathfrak{G})$ and $(\mathrm{H}_c)_\delta$.

Using (11), we have that

$$\Omega_{c+k}({}_{c+k}\mathrm{P}_c \otimes_{\mathrm{A}_c} L) = {}_{-(c+k)-1}\mathrm{Q}_{-c-1} \otimes_{\mathrm{A}_{-c-1}} \Omega_c(L).$$

By (14), we have a filtered isomorphism

$$\Omega_{c+k}({}_{c+k}\mathrm{P}_c \otimes_{\mathrm{A}_c} L) \cong \mathcal{D}_{-c}(\mathfrak{G})^{\det^{-k}} \otimes_{\mathrm{A}_{-c-1}} M^{\tau_{-c}(\overline{\mathfrak{g}})}. \tag{15}$$

Consider the tensor product filtration on $\mathcal{D}_{-c}(\mathfrak{G})^{\det^{-k}} \otimes_{\mathrm{A}_{-c-1}} M^{\tau_{-c}(\mathfrak{g})}$ and the sub-filtration on $M^{\tau_{-c-k}(\mathfrak{g})}$. The homomorphism ϕ_M^k is filtered, i.e. for any $i \geq 0$,

$$\phi_M^k F_i \left(\mathcal{D}_{-c}(\mathfrak{G})^{\det^{-k}} \otimes_{\mathrm{A}_{-c-1}} M^{\tau_{-c}(\overline{\mathfrak{g}})}\right) \subset F_i(M^{\tau_{-c-k}(\overline{\mathfrak{g}})}).$$

Given (15), the lemma follows. □

2.6.2. *A descent functor.* Let $F\mathscr{C}(\mathfrak{G})$ be the category whose objects are pairs $(\mathbf{M}, F_\bullet)$, where $\mathbf{M}$ is a mirabolic D-module and F is a good filtration on $\mathbf{M}$. We define a descent functor

$$\begin{aligned} \Psi_c : F\mathscr{C}^G(\mathfrak{G}) &\to \mathrm{Coh}(\mathrm{Hilb}^n) \\ (\mathbf{M}, F_\bullet) &\mapsto \mathrm{Proj} \bigoplus_{\ell \geq 0} \Gamma(\widetilde{\mathrm{Hilb}}^n, \widetilde{\mathrm{gr}}^F \mathbf{M})^{\tau_{-c-\ell}(\overline{\mathfrak{g}})} \end{aligned} \tag{16}$$

whose essential image lands in $\mathrm{Coh}(\mathrm{Hilb}_1^n)$ where $\mathrm{Hilb}_1^n \subset \mathrm{Hilb}^n$ is the preimage of $\{(x,0) \in (\mathbb{C}^n)^2\}/S_n$ under the Hilbert-Chow map.

2.6.3. *Gradings.* Let $(x_{ij})_{1\leq i,j\leq n}$ and $(\partial_{x_{ij}})_{1\leq i,j\leq n}$ be the dual bases of $\overline{\mathfrak{g}}^*$ and $\overline{\mathfrak{g}}$ respectively. Let $\{x_i\}$ and $\{y_i\}$ be dual bases of $\mathfrak{h}^*$ and $\mathfrak{h}$ respectively.

Define

$$\begin{aligned} \mathcal{D}(\mathfrak{G}) \ni \widetilde{h} :=& \frac{1}{2} \sum_{1\leq i,j\leq n} (x_{ij}\partial_{x_{ij}} + \partial_{x_{ij}} x_{ij}) \\ \mathrm{H}_c \ni \mathrm{h}_c :=& \Omega_{-c-1}(\mathscr{H}_{-c}(\widetilde{h})) = \frac{1}{2} \sum_{i=1}^n (x_i y_i + y_i x_i). \end{aligned} \tag{17}$$

Let $F\mathcal{O}(\mathrm{A}_c)$ be the category whose objects are pairs $(L, F_\bullet)$ where L is an object in $\mathcal{O}(\mathrm{A}_c)$ and $F_\bullet$ is a good filtration on L. For any $(\mathbf{M}, F_\bullet) \in F\mathscr{C}^G(\mathfrak{G})$, resp. $(L, F_\bullet) \in F\mathcal{O}(\mathrm{A}_c)$, the action of $\widetilde{h}$,

resp. h_c, is semisimple and induces a $\mathbb{Z}$-grading. This grading together with the filtration induces a $\mathbb{C}^*\times\mathbb{C}^*$-equivariant structure on $\Psi_c(\mathbf{M})$, resp. $GS(L)$.

On the other hand, the scalar action of $\mathbb{C}^*\times\mathbb{C}^*$ on $\mathbb{C}^2$ induces an $\mathbb{C}^*\times\mathbb{C}^*$-action on Hilb^n. We have the following correspondence:

$$\text{filtration grading} \leftrightarrow \left[C^*\hookrightarrow \mathbb{C}^*\times\mathbb{C}^*:, z\mapsto (1,z)\right] \tag{18}$$

$$\mathrm{h}_c\text{-grading} \leftrightarrow \left[C^*\hookrightarrow \mathbb{C}^*\times\mathbb{C}^*: z\mapsto (z,z^{-1})\right]. \tag{19}$$

2.6.4. *Commutativity of the diagram.* Consider the following diagram:

$$\begin{array}{ccc} & F\mathscr{C}(\mathfrak{G}) & \\ {\scriptstyle \Omega_{-c-1}\circ\mathbb{H}_{-c}}\swarrow & & \searrow{\scriptstyle \Psi_c} \\ F\mathcal{O}(\mathrm{A}_c) & \xrightarrow{\quad GS\quad} & \mathrm{Coh}^{\mathbb{C}^*\times\mathbb{C}^*}(\mathrm{Hilb}^n) \end{array} \tag{20}$$

Proposition 2.7. *Let $(\mathbf{M},F_\bullet)\in F\mathscr{C}^G(\mathfrak{G})$ and $(M,F_\bullet)=\Gamma(\mathfrak{G},(\mathbf{M},F_\bullet))$. Suppose that $-c-k$ satisfies (4) for all $k\geq 1$. Then the following are equivalent:*

(a) $\Psi_c(\mathbf{M})\cong GS\circ\Omega_{-c-1}\circ\mathbb{H}_{-c}(\mathbf{M})$.
(b) ϕ_M^ℓ (defined in (12)) is a filtered isomorphism for all $\ell\gg 0$.
(c) ϕ_M^ℓ is a filtered isomorphism for all $\ell\geq 0$.

Proof. Let $L=\Omega_{-c-1}\circ\mathbb{H}_c(M)$. By definition,

$$GS(L)=\mathrm{Proj}(\bigoplus_{\ell\geq 0}\mathrm{gr}S_{c,\ell}L).$$

On the other hand,

$$\Psi_c(\mathbf{M})=\mathrm{Proj}(\bigoplus_{\ell\geq 0}\Gamma(\widetilde{\mathrm{Hilb}},\widetilde{\mathrm{gr}}^F\mathbf{M})^{\tau_{-c-\ell}(\overline{\mathfrak{g}})}).$$

Therefore, the equality $GS(L)=\Psi_c(\mathbf{M})$ is equivalent to

$$\Gamma(\widetilde{\mathrm{Hilb}},\widetilde{\mathrm{gr}}^F\mathbf{M})^{\tau_{-c-\ell}(\overline{\mathfrak{g}})}\cong \mathrm{gr}^T S_{c,\ell}L,\quad \forall \ell\gg 0.$$

By [GGS09, Proposition 7.4], since $\widetilde{\mathrm{Hilb}}$ is the semistable locus with respect to the character det of $\overline{G}$, there is

$$\Gamma(\widetilde{\mathrm{Hilb}},\widetilde{\mathrm{gr}}^F\mathbf{M})^{\tau_{-c-\ell}(\overline{\mathfrak{g}})}=\Gamma(T^*\mathfrak{G},\widetilde{\mathrm{gr}}^F\mathbf{M})^{\tau_{-c-\ell}(\overline{\mathfrak{g}})}=\mathrm{gr}^F\Gamma(\mathfrak{G},\mathbf{M})^{\tau_{-c-\ell}(\overline{\mathfrak{g}})},\quad \text{for } \ell\gg 0.$$

Moreover, $\Omega_{-c-\ell-1}$ preserves filtration, i.e. $\mathrm{gr}^F\Omega_{-c-\ell-1}M^{\tau_{-c-k}(\overline{\mathfrak{g}})}\cong \mathrm{gr}^F M^{\tau_{-c-\ell}(\overline{\mathfrak{g}})}$. Given the discussion around (13), we conclude that (a) is equivalent to (b).

As for the implication (b) $\Rightarrow$ (c), for any $k\geq 0$, take $\ell\gg 0$ and consider

$$\begin{array}{ccc} \mathcal{D}_{-c}(\mathfrak{G})^{\det^{-\ell}}\otimes_{\mathrm{A}_{-c-k-1}}\mathcal{D}_{-c}(\mathfrak{G})^{\det^{-k}}\otimes_{\mathrm{A}_{-c-1}}M^{\tau_{-c}(\overline{\mathfrak{g}})} & \xrightarrow{id\otimes\phi_M^k} & \mathcal{D}_{-c}(\mathfrak{G})^{\det^{-\ell}}\otimes_{\mathrm{A}_{-c-k-1}}M^{\tau_{-c-k}(\overline{\mathfrak{g}})} \\ \downarrow{\scriptstyle \mathrm{mul}\otimes id} & & \downarrow{\scriptstyle \phi_M^\ell} \\ \mathcal{D}_{-c}(\mathfrak{G})^{\det^{-\ell-k}}\otimes_{\mathrm{A}_{-c-1}}M^{\tau_{-c}(\overline{\mathfrak{g}})} & \xrightarrow{\phi_M^{\ell+k}} & M^{\tau_{-c-\ell-k}(\overline{\mathfrak{g}})} \end{array}$$

Here the map $\mathrm{mul}:\mathcal{D}_{-c}(\mathfrak{G})^{\det^{-\ell}}\otimes_{\mathrm{A}_{-c-k-1}}\mathcal{D}_{-c}(\mathfrak{G})^{\det^{-k}}\to\mathcal{D}_{-c}(\mathfrak{G})^{\det^{-\ell-k}}$ is defined by multiplication. By [GGS09, Lemma 5.2(2)], mul is a filtered isomorphism. By assumption, $\phi_M^{\ell+k}$, ϕ_M^ℓ are filtered isomorphisms. As a result, ϕ_M^k is also a filtered isomorphism. □

3. Hodge filtrations on cuspidal D-modules

In Saito's theory [Sai88, Sai90], the structure of a Hodge module on a smooth algebraic variety X consists of the information of a triple $(M, F_\bullet, V_{\mathbb{Q}})$: M is a regular holonomic D_X-module, $F_\bullet$ is a good filtration on M called the Hodge filtration and $V_{\mathbb{Q}}$ is a $\mathbb{Q}$-perverse sheaf such that M is mapped to $V_{\mathbb{Q}} \otimes_{\mathbb{Q}} \mathbb{C}$ under the Riemann-Hilbert correspondence. These data satisfy a series of compatibility, defined inductively with respect to dimensions using nearby and vanishing cycles. We refer the readers to [Sch] for an overview of the precise definition.

Suppose Z is an irreducible closed subvariety of X. For any Hodge module M on X strictly supported on Z (i.e. no quotients or sub-objects supported on a proper subset of Z), there exists an open subvariety $U \subset Z$ such that the restriction of M to U is a variation of Hodge structure (VHS), which uniquely determines the original module M. Conversely, the structure theorem of Saito [Sai90, Theorem 3.21] ensures that any polarizable VHS on U extends uniquely to a Hodge module on X strictly supported on Z.

The cuspidal D-module $\mathbf{N}_c$, resp. $\overline{\mathbf{N}}_c$ has the structure of a Hodge module strictly supported on $\mathscr{N}$, resp. $\mathscr{N} \times V$. Our goal is to compute the associated graded of $\mathbf{N}_c$ with respect to the Hodge filtration. To do so, we first fix a Borel subgroup $B \subset G$ with Lie algebra $\mathfrak{b} \subset \mathfrak{g}$ of upper triangular matrices and study the Hodge filtration on the restriction of $\mathbf{N}_c$ to $\mathfrak{b}$ explicitly.

We adapt a naive notion of $\mathbb{C}$-Hodge modules, in the form of $(M, F_\bullet)$ by forgetting the data of $V_{\mathbb{Q}}$ in a Hodge module, as in [Sai22, Remark 2.6].

Below, there are various filtrations, but only one on each D-module and $\widetilde{\mathrm{gr}}$ indicates taking associated graded with respect to that filtration. To ease notations, all of the filtrations will be denoted by F. On $\mathcal{D}_X$ we will only consider the order filtration.

3.1. Deligne's canonical lattice.

Theorem 3.1. *([Del70],[HTT08, Theorem 5.2.17]) Let D be a simple normal crossing (SNC) divisor in X and $j : U := X \setminus D \hookrightarrow X$ be the open embedding. Suppose $\mathcal{F}$ is an integrable connection on U. There is a unique locally free $\mathcal{O}_X$-module $\mathcal{F}^{>-1}$ extending $\mathcal{F}$, known as Deligne's canonical lattice, such that the following hold:*

(1) $\mathcal{O}[D] \otimes \mathcal{F}^{>-1}$ has the structure of a meromorphic extension of $\mathcal{F}$ with a logarithmic pole along D with respect to $\mathcal{F}^{>-1}$.
(2) Along any irreducible component of D, the residue has eigenvalues in $(-1, 0]$.

Here $\mathcal{O}[D]$ denotes the sheaf of rational functions on X with possible poles only along D. We refer readers to [HTT08, 5.2.2] for definitions for the other terminologies involved.

Keep the setting of Theorem 3.1 and further assume that $\mathcal{F}$ is a polarizable VHS with a Hodge filtration F. The Hodge module on X extended from $\mathcal{F}$, whose existence is implied by the structure theorem, has underlying D-module being $i_{!*}\mathcal{F}$, which satisfies that $i_{!*}\mathcal{F} = \mathcal{D}_X \cdot \mathcal{F}^{>-1}$. According to [Sai90, 3.b] (cf. [Pop18, 4.4]), the lattice $\mathcal{F}^{>-1}$ is filtered by locally free $\mathcal{O}_X$-submodules:

$$F_k\mathcal{F}^{>-1} = \mathcal{F}^{>-1} \cap j_*F_k\mathcal{F},$$

and the Hodge filtration on $i_{!*}\mathcal{F}$ is defined by

$$F_k(i_{!*}\mathcal{F}) = \sum_i F_i\mathcal{D}_X F_{k-i}\mathcal{F}^{>-1}.$$

3.2. **Hodge filtrations restricted to $\mathfrak{b}$.** Let $\mathfrak{n} = [\mathfrak{b}, \mathfrak{b}]$ and $\mathfrak{n}_r = \mathfrak{n} \cap \mathscr{N}_r$. We give $\mathfrak{n} \times V$ coordinates by

$$\begin{pmatrix} 0 & x_1 & * & * & * & * \\ 0 & 0 & x_2 & * & * & * \\ & & & \cdots & x_{n-2} & * \\ 0 & 0 & 0 & \cdots & 0 & x_{n-1} \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ \vdots \\ \\ \vdots \\ v_n \end{pmatrix} \tag{21}$$

The restriction of the function s (cf. (7)) to $\mathfrak{n} \times V$ equals $x_1 x_2^2 \cdots x_{n-1}^{n-1} v_n^n$. Meanwhile, the complement D of $\mathfrak{n}_r$ in $\mathfrak{n}$ is a SNC divisor defined by the vanishing of $x_1 x_2 \cdots x_{n-1}$.

Take $c = \frac{m}{n}$ for a positive integer m coprime to n. Recall the $\mathcal{D}_U$-module $\mathcal{E}_c$ from Section 2.3.2. The restriction of $\mathcal{E}_c$ to $U_0 := U \cap (\mathfrak{n} \times V)$ is generated by the B-invariant horizontal section

$$s_0^c := s^c|_{U_0} = x_1^c x_2^{2c} \cdots x_{n-1}^{(n-1)c} v_n^m. \tag{22}$$

The integrable connection $\mathcal{F}_c^\circ := \mathcal{F}_c|_{\mathfrak{n}_r}$ is also generated by the B-invariant section s_0^c: the extra factor v_n^m is necessary for s_0^c to be T-invariant, since $(t_1, \cdots, t_n) \in T$ acts by

$$t_1^c t_2^{-c+2c} \cdots t_{n-1}^{(n-2)c-(n-1)c} t_n^{-(n-1)c+nc} = (t_1 \cdots t_n)^c.$$

The Hodge filtration F on the VHS $\mathcal{F}_c^\circ$ is defined by $F_k \mathcal{F}_c^\circ = \mathcal{F}_c^\circ$ for $k \geq 0$ and 0 otherwise.

Write $i_0 : \mathfrak{n}_r \hookrightarrow \mathfrak{n}$ and $\mathbf{L}_c^\circ := (i_0)_{!*} \mathcal{F}_c^\circ$, which is isomorphic to $(i_0)_\dagger \mathcal{F}_c^\circ$. We apply Section 3.1 to studying $\mathbf{L}_c^\circ$.

Let

$$\lceil s_0^c \rceil := x_1^{\lceil c \rceil} x_2^{\lceil 2c \rceil} \ldots x_{n-1}^{\lceil (n-1)c \rceil} v^m. \tag{23}$$

We compute that

$$x_i \partial_{x_i} (s_0^c \lceil s_0^c \rceil^{-1}) = (ic - \lceil ic \rceil) s_0^c \lceil s_0^c \rceil^{-1}, \quad i = 1, \ldots, n-1.$$

Since $\mathcal{O}[D] \otimes_{\mathcal{O}_\mathfrak{n}} \mathcal{O}_\mathfrak{n} \lceil s_0^c \rceil^{-1} s_0^c \cong \mathbf{L}_c^\circ$ and $ic - \lceil ic \rceil \in (-1, 0]$ for all i, we see that $(\mathcal{F}_c^\circ)^{>-1} = \mathcal{O}_\mathfrak{n} \lceil s_0^c \rceil^{-1} s_0^c$.

The lattice $(\mathcal{F}_c^\circ)^{>-1}$ is filtered by

$$F_k (\mathcal{F}_c^\circ)^{>-1} = \begin{cases} (\mathcal{F}_c^\circ)^{>-1} \cap (i_0)_* F_k \mathcal{F}_c^\circ = \mathcal{O}_\mathfrak{n} \lceil s_0^c \rceil^{-1} s_0^c & k \geq 0 \\ 0 & k < 0 \end{cases}$$

and the Hodge filtration on $\mathbf{L}_c^\circ$ is thus

$$F_k \mathbf{L}_c^\circ = \sum F_i \mathcal{D}_\mathfrak{n} F_{k-i} (\mathbf{L}_c^\circ)^{>-1} = \begin{cases} (F_k \mathcal{D}_\mathfrak{n}) \cdot \lceil s_0^c \rceil^{-1} s_0^c & k \geq 0 \\ 0 & k < 0 \end{cases} \tag{24}$$

Let $i_\mathfrak{n} : \mathfrak{n} \to \mathfrak{g}$ and $\mathcal{D}_{\mathfrak{g} \leftarrow \mathfrak{n}} := i_\mathfrak{n}^* \mathcal{D}_\mathfrak{g} \otimes_{\mathcal{O}_\mathfrak{n}} \omega_{\mathfrak{n}/\mathfrak{g}}$. The pushforward of the Hodge module $(\mathbf{L}_c^\circ, F_\bullet)$ has the underlying $\mathcal{D}_\mathfrak{g}$-module

$$\mathbf{L}_c = (i_\mathfrak{n})_\dagger \mathbf{L}_c^\circ = i_* \left(\mathcal{D}_{\mathfrak{g} \leftarrow \mathfrak{n}} \otimes_{\mathcal{D}_\mathfrak{n}} \mathbf{L}_c^\circ \right)$$

with the Hodge filtration defined by ([Pop18, 1.5])

$$F_k \mathbf{L}_c = \mathrm{Im} \left(\left(\sum_q F_j \mathcal{D}_{\mathfrak{g} \leftarrow \mathfrak{n}} \otimes F_{k-j}^D (\mathbf{L}_c^\circ) \right) \to (i_\mathfrak{n})_\dagger \mathbf{L}_c^\circ \right). \tag{25}$$

Let $\overline{\mathbf{L}}_c$ be the minimal extension of $\mathcal{E}_c|_{U_0}$ to $\mathfrak{G}$. One can run the same procedure to describe a similar Hodge module structure on $\overline{\mathbf{L}}_c$.

3.3. **Associated graded of L.** Recall the standard $\mathfrak{sl}_2$-triple in $\mathfrak{g}$:

$$\underline{e} = \begin{pmatrix} 0 & 1 & \cdots & 0 & 0 \\ & & \cdots & & \\ 0 & 0 & \cdots & 1 & 0 \\ 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & \cdots & 0 & 0 \end{pmatrix} \quad \underline{f} = \begin{pmatrix} 0 & 0 & \cdots & 0 & 0 \\ n-1 & 0 & \cdots & 0 & 0 \\ 0 & 2(n-2) & \cdots & 0 & 0 \\ & & \cdots & & \\ 0 & 0 & \cdots & 1-n & 0 \end{pmatrix} \quad \underline{h} = [\underline{e}, \underline{f}] \tag{26}$$

Define the Hessenberg space $\mathfrak{m} = \overline{B \cdot (\underline{f} + \mathrm{stab}_{\mathfrak{g}}(\underline{e}))}$, which is the sum of $\mathfrak{m}$ and the span of negative simple root spaces.

Remark 3.1. The affine space $\underline{f} + \mathrm{stab}_{\mathfrak{g}}(\underline{e})$ is the Kostant slice. The Whittaker reduction of $G \times^B \mathfrak{m}$ is the compactified regular centralizer [Bal23].

Under the coordinates (21) and

$$\begin{pmatrix} * & * & \cdots & * & * & * \\ y_1 & * & \cdots & * & * & * \\ 0 & y_2 & \cdots & * & * & * \\ & & \cdots & & & \\ 0 & 0 & \cdots & y_{n-2} & * & * \\ 0 & 0 & \cdots & 0 & y_{n-1} & * \end{pmatrix} \tag{27}$$

we define

$$\mathfrak{Y}_0 := \{(x, y) \in \mathfrak{n} \times \mathfrak{m} | x \in \mathfrak{n}, y \in \mathfrak{m}, x_i y_i = 0, 1 \leq i \leq n-1\}. \tag{28}$$

Let $i_{\mathfrak{Y}_0} : \mathfrak{Y}_0 \to T^*\mathfrak{g}$ be the embedding.

Note the identification

$$\Gamma(\mathfrak{g}, \mathbf{L}_c) \cong \frac{\mathcal{D}(\mathfrak{g})}{\mathcal{D}(\mathfrak{g}) \cdot \mathcal{O}(\mathfrak{b}_-) + \sum_{i=1}^{n-1} \mathcal{D}(\mathfrak{g})(x_i \partial_{x_i} - ic) + \mathcal{D}(\mathfrak{g}) \cdot S([\mathfrak{n}, \mathfrak{n}])}.$$

Given (24) and (25), we see that $\widetilde{\mathrm{gr}}\, \mathbf{L} \cong (i_{\mathfrak{Y}_0})_* \mathcal{O}_{\mathfrak{Y}_0}$ with respect to the Hodge filtration.

Recall the shifted embedding $\tau_{-c} : \overline{\mathfrak{g}} \to \mathcal{D}(\mathfrak{G}), x \mapsto \tau(x) + c\mathrm{tr}(x)$ from (6). Similarly define τ' and $\tau'_{-c} : \overline{\mathfrak{g}} \to \mathcal{D}(\mathfrak{g}), x \mapsto \tau'(x) + c\mathrm{tr}(x)$. Although the D-modules $\mathbf{L}_c$ and $\overline{\mathbf{L}}_c$ are only B-equivariant but not $\overline{B}$-equivariant, the coherent sheaves $\widetilde{\mathrm{gr}}\, \mathbf{L}_c$ and $\widetilde{\mathrm{gr}}\, \overline{\mathbf{L}}_c$ are naturally $\overline{B}$-equivariant:

Lemma-Construction 3.1. On $\widetilde{\mathrm{gr}}\, \mathbf{L}_c$, resp. $\widetilde{\mathrm{gr}}\, \overline{\mathbf{L}}_c$, the $\overline{\mathfrak{b}}$-action via the embedding τ'_{-c}, resp. τ_{-c} integrates into a $\overline{B}$-action.

Proof. Any $z \in \overline{\mathfrak{b}}$ acts via τ on s_0^c through $\overline{\mathfrak{h}}$ by $-c(z_1 + \cdots + z_n)$ where z_i's denote the diagonals of z. Thus $\tau_{-c}(z) s_0^c = 0$ for all $z \in \overline{\mathfrak{b}}$. Therefore, $\overline{\mathfrak{h}}$ acts via τ_{-c} on $\mathbf{L}_c \cong \mathcal{D}(\mathfrak{g}) s_0^c$ and hence on $\widetilde{\mathrm{gr}}\, \mathbf{L}_c$ by integral weights. It follows that the $\tau_{-c}(\overline{\mathfrak{h}})$ action on the coherent sheaf $\widetilde{\mathrm{gr}}\, \mathbf{L}_c$ integrates to a $\overline{T}$ action, compactible with the B-equivariance passed from $\mathbf{L}_c$. The same computation applies to $\widetilde{\mathrm{gr}}\, \overline{\mathbf{L}}_c$ and the lemma follows. □

This $\overline{B}$-equivariance is consistent with the definition of Ψ_c (16). We proceed to describe the $\overline{B}$-equivariance of $\widetilde{\mathrm{gr}}\, \mathbf{L}_c$ and $\widetilde{\mathrm{gr}}\, \overline{\mathbf{L}}_c$ under Construction 3.1.

The following two weights will be important

$$\lceil \mu_c \rceil = (\lceil \mu_c \rceil(1), \cdots, \lceil \mu_c \rceil(n)) := (\lceil c \rceil, \lceil 2c \rceil - \lceil c \rceil, \ldots, \lceil nc \rceil - \lceil (n-1)c \rceil) \tag{29}$$

$$\lfloor \mu_c \rfloor = (\lfloor \mu_c \rfloor(1), \cdots, \lfloor \mu_c \rfloor(n)) := (\lfloor c \rfloor, \lfloor 2c \rfloor - \lfloor c \rfloor, \ldots, \lfloor nc \rfloor - \lfloor (n-1)c \rfloor) \tag{30}$$

and they satisfy $\lfloor \mu_c \rfloor = w_0 \lceil \mu_c \rceil$ where w_0 is the longest element in W.

Let $i_V : V \hookrightarrow T^*V$ be the zero section. Endow $\mathcal{O}_{\mathcal{Y}_0}$, resp. $\mathcal{O}_{\mathcal{Y}_0} \boxtimes (i_V)_*\mathcal{O}_V$ with the trivial $\overline{B}$-equivariant structure. For any character λ of $\overline{T}$, let $\mathbb{C}_\lambda$ be the 1-dimensional representation of $\overline{B}$ associated to λ. Let α be the weight associated to the relative canonical bundle $\omega_{\mathfrak{n}/\mathfrak{g}}$.

Lemma 3.2. *Under Lemma-Construction 3.1, the following statements hold:*

(1) As $\overline{B}$-equivariant $\mathcal{O}_{T^*\mathfrak{g}}$-modules, $\widetilde{\mathrm{gr}}\ \mathbf{L}_c = (i_{\mathcal{Y}_0})_*\mathcal{O}_{\mathcal{Y}_0} \otimes \mathbb{C}_{\lceil \mu_c \rceil + \alpha}$.

(2) As $\overline{B}$-equivariant $\mathcal{O}_{T^*\mathfrak{G}}$-modules, $\widetilde{\mathrm{gr}}\ \overline{\mathbf{L}}_c = ((i_{\mathcal{Y}_0})_*\mathcal{O}_{\mathcal{Y}_0} \boxtimes (i_V)_*\mathcal{O}_V) \otimes \mathbb{C}_{\lceil \mu_c \rceil + \alpha}$.

Proof. We have seen that $\widetilde{\mathrm{gr}}\ \mathbf{L}_c \cong (i_{\mathcal{Y}_0})_*\mathcal{O}_{\mathcal{Y}_0}$. Moreover, the first non-vanishing filtered piece of $\mathrm{gr}\ \mathbf{L}_c$ equals $\omega_{\mathfrak{n}/\mathfrak{g}} \otimes \mathbb{C}\lceil s_0^c \rceil^{-1} s_0^c$. The lemma follows from that $\overline{B}$-acts on $\lceil s_0^c \rceil^{-1}$ (23) (via (5)) exactly by the character $\lceil \mu_c \rceil$ through $\overline{B} \twoheadrightarrow \overline{T}$. (1) follows and verbatim for (2). □

3.4. **Functors on equivariant Hodge modules.** We state two important results on Hodge modules, which allow us to describe $\widetilde{\mathrm{gr}}\ \mathbf{N}_c$ in terms of $\widetilde{\mathrm{gr}}\ \mathbf{L}_c$.

In this section, let G be an affine algebraic group and X be a smooth variety with a G-action. We denote the abelian category of G-equivariant Hodge modules on X by $\mathrm{HM}^G(X)$.

3.4.1. *Inductions.* Let H be a closed subgroup of G, which acts on the product $G \times X$ by $h \cdot (g, x) = (gh^{-1}, hx)$. Let $G \times^H X$ be the quotient of this action. Consider the diagram:

$$\begin{array}{ccc} G \times X & \xrightarrow{\pi} & G \times^H X \\ \downarrow{\scriptstyle pr} & & \downarrow{\scriptstyle a} \\ X & & X \end{array}$$

where pr is the second projection, π is the quotient map and $a : \overline{(g, x)} \mapsto gx$.

For any H-equivariant $\mathcal{D}_X$-module $\mathcal{F}$, there is a unique G-equivariant $\mathcal{D}_{G\times^H X}$-module $\mathcal{E}$ such that $pr^\dagger \mathcal{F} \cong \pi^\dagger \mathcal{E}$ [BL94]. We denote $\mathcal{E} = \mathrm{Ind}_H^G \mathcal{F}$ and $\widetilde{\mathrm{Ind}}_H^G := a_\dagger \circ \mathrm{Ind}_H^G$.

Similarly, for any H-equivariant coherent $\mathcal{O}_X$-module $\mathcal{F}$, let $\mathrm{ind}_H^G \mathcal{F}$ denote the corresponding G-equivariant $\mathcal{O}_{G\times^H X}$-module.

For any map $f : X \to Y$, we introduce the associated Lagrangian correspondence: ρ_f is the co-differential of f and ϖ_f is the natural projection:

$$T^*X \xleftarrow{\rho_f} X \times_Y T^*Y \xrightarrow{\varpi_f} T^*Y \tag{31}$$

Combining the cases when $f = \pi$ and $f = pr$, we obtain a commutative diagram:

$$\begin{array}{ccccc} T^*(G \times X) & \xleftarrow{\rho_{pr}} & G \times T^*X & \xrightarrow{\varpi_{pr}} & T^*X \\ \uparrow{\scriptstyle \rho_\pi} & & & \searrow{\scriptstyle q} & \\ T^*(G \times^H X) \times_{G\times^H X} (G \times X) & \xrightarrow{\varpi_\pi} & T^*(G \times^H X) & \xleftarrow{s} & G \times^H T^*X \end{array} \tag{32}$$

Here q is the quotient map and$s$ is the natural embedding.

Proposition 3.3. *For any $\mathbf{M} \in \mathrm{HM}^G(G \times^H X)$, we have $SS(\mathbf{M}) \subset s(G \times^H T^*X)$ and the following diagram commutes:*

$$\begin{array}{ccc} \mathrm{HM}^H(X) & \xrightarrow[\sim]{\mathrm{Ind}_H^G} & \mathrm{HM}^G(G \times^H X) \\ \downarrow{\scriptstyle \widetilde{\mathrm{gr}}} & & \downarrow{\scriptstyle \widetilde{\mathrm{gr}}} \\ \mathrm{Coh}^H(T^*X) & \xrightarrow[\sim]{\mathrm{ind}_H^G} & \mathrm{Coh}^G(G \times^H T^*X) \end{array}$$

Proof. By [Ach, Theorem 6.2], the arrow Ind_H^G on the top is an equivalence. It remains to show the essential image of $\mathrm{HM}^G(G \times^H X)$ under $\widetilde{\mathrm{gr}}$ is as desired and the diagram is commutative.

Suppose $\mathbf{M} = \mathrm{Ind}_H^G \mathbf{L}$ for $\mathbf{L} \in \mathrm{HM}^H(X)$. By definition, $\pi^\dagger \mathbf{M} = pr^\dagger \mathbf{L}$ as G-equivariant $\mathcal{D}_{G\times X}$-modules. Taking associated graded of both sides and using [Kas03, Theorem 4.7], we obtain an isomorphism in $\mathrm{Coh}^G(T^*(G \times X))$

$$(\rho_\pi)_* \varpi_\pi^* \widetilde{\mathrm{gr}}\ \mathbf{M} \cong (\rho_{pr})_* \varpi_{pr}^* \widetilde{\mathrm{gr}}\ \mathbf{L}. \tag{33}$$

In particular, we see $SS(\mathbf{M}) \subset \varpi_\pi \rho_\pi^{-1} \rho_{pr}(G \times T^*X) \subset s(G \times^H T^*X)$.

Given the diagram (32), the identity (33) implies $\varpi_{pr}^* \widetilde{\mathrm{gr}}\ \mathbf{L} = q^* \widetilde{\mathrm{gr}}\ \mathbf{M}$. Therefore $\mathrm{ind}_B^G \widetilde{\mathrm{gr}}\ \mathbf{L} = \widetilde{\mathrm{gr}}\ \mathbf{M}$ and the diagram in the statement commutes. □

3.4.2. *Associated graded of a pushforward.* Let $p : X \to Y$ be a projective morphism between two smooth varieties and $M \in \mathrm{HM}^G(X)$. By the directed image theorem of Saito ([Sai90, Theorem 2.14]), $R^i p_\dagger(M) \in \mathrm{HM}(Y)$. Furthermore, by [Ach, Theorem 8.5], $R^i p_\dagger(M) \in \mathrm{HM}^G(Y)$. Let $\widetilde{\omega_{X/Y}}$ denote the pullback of the relative canonical bundle $\omega_{X/Y} = \omega_X \otimes p^* \omega_Y^{-1}$ to $X \times_Y T^*Y$.

Theorem 3.4. *(*[Lau83, 2.3.2], [Sch, 28]*)The diagram below commutes.*

$$\begin{array}{ccc} \mathrm{HM}^G(X) & \xrightarrow{R^i p_\dagger} & \mathrm{HM}^G(Y) \\ \downarrow{\scriptstyle \widetilde{\mathrm{gr}}} & & \downarrow{\scriptstyle \widetilde{\mathrm{gr}}} \\ \mathrm{Coh}^G(T^*X) & \xrightarrow{R^i(\varpi_p)_*(\widetilde{\omega_{X/Y}} \otimes L\rho_p^*(-))} & \mathrm{Coh}^G(T^*Y) \end{array}$$

Remark 3.2. This result does not hold in general if $\widetilde{\mathrm{gr}}$ is taken with respect to an arbitrary good filtration.

3.5. **Springer cuspidal D-modules.** We apply the two results in the last section to computing $\widetilde{\mathrm{gr}}\ \mathbf{N}_c$.

Let $\mathcal{B}$ denote the flag variety. Since every regular nilpotent x is contained in a unique Borel subalgebra, there are embeddings of $\mathcal{N}_r$ into $\mathcal{B} \times \mathfrak{g}$ and of U into $\mathcal{B} \times \mathfrak{G}$ such that the following diagrams commute:

$$\begin{array}{ccc} \mathcal{N}_r & \hookrightarrow & \mathfrak{g} \\ & {\scriptstyle \widetilde{i}}\searrow & \uparrow{\scriptstyle p} \\ & & \mathcal{B} \times \mathfrak{g} \end{array} \qquad \begin{array}{ccc} U & \hookrightarrow & \mathfrak{G} \\ & \searrow & \uparrow \\ & & \mathcal{B} \times \mathfrak{G} \end{array}$$

Recall we defined D-modules $\mathcal{E}_c$ and $\mathcal{F}_c$ in Section 2.3.2.

Definition 3.2. The Springer cuspidal $\mathcal{D}_{\mathfrak{g}}$-module $\mathbf{M}_c$ is the minimal extension of $\mathcal{F}_c$ to $\mathcal{B} \times \mathfrak{g}$. The Springer cuspidal $\mathcal{D}_{\mathfrak{G}}$-module $\overline{\mathbf{M}}_c$ is the minimal extension of $\mathcal{E}_c$ to $\mathcal{B} \times \mathfrak{G}$.

The local system $\mathcal{F}_c$ is clean with respect to the inclusion $\widetilde{i}$, i.e. $\mathbf{M}_c = \widetilde{i}_\dagger \mathcal{F}$. Therefore by functoriality, $p_\dagger \mathbf{M}_c = \mathbf{N}_c$. Similar to Lemma 2.3, we have that $\overline{\mathbf{M}}_c = \mathbf{M}_c \boxtimes \mathcal{O}_V$.

Lemma 3.5.

(1) The G-equivariant $\mathcal{D}_{\mathcal{B}\times\mathfrak{g}}$-module underlying $\mathrm{Ind}_B^G(\mathbf{L}, F_\bullet)$ is $\mathbf{M}_c$.
(2) The G-equivariant $\mathcal{D}_{\mathcal{B}\times\mathfrak{G}}$-module underlying $\mathrm{Ind}_B^G(\overline{\mathbf{L}}, F_\bullet)$ is $\overline{\mathbf{M}}_c$.

Proof. We only show (1) and the same argument applies to (2) since $\overline{\mathbf{L}} \cong \mathbf{L} \boxtimes \mathcal{O}_V$. Consider the following Cartesian diagram

$$\begin{array}{ccccccc} \mathfrak{n}_r & \xleftarrow{p_0} & G \times \mathfrak{n}_r & \xrightarrow{\pi_0} & G \times^B \mathfrak{n}_r & = & \mathscr{N}_r \\ \downarrow{\scriptstyle i_{0,r}} & & \downarrow{\scriptstyle i_r} & & & & \downarrow{\scriptstyle i} \\ \mathfrak{g} & \xleftarrow{p} & G \times \mathfrak{g} & & \xrightarrow{\quad\pi\quad} & & \mathcal{B} \times \mathfrak{g} \end{array}$$

Since $\mathcal{E}_c = \mathrm{Ind}_B^G \mathbf{L}_0$, using base change twice, we have

$$\pi^\dagger i_\dagger(\mathcal{E}_c) = (i_r)_\dagger \pi_0^\dagger(\mathcal{E}_c) = (i_r)_\dagger p_0^\dagger(\mathcal{F}_c) = p^\dagger (i_{0,r})_\dagger(\mathcal{F}_c)$$

which proves the lemma. □

Recall that $SS(\mathbf{L}) = \mathfrak{Y}_0$ for $\mathfrak{Y}_0$ as defined in (28). Also recall the coordinates from (21) and (27). When $x \in \mathfrak{n}$ and $y \in \mathfrak{m}$, we have $[x, y] \in \mathfrak{b}$. Moreover, the diagonals of $[x, y]$ equal

$$x_1 y_1,\ x_2 y_2 - x_1 y_1, \ldots, x_{n-1} y_{n-1} - x_{n-2} y_{n-2}, -x_{n-1} y_{n-1}.$$

As a result, $x_i y_i = 0$, $1 \le i \le n-1$ if and only if $[x, y] \in \mathfrak{n}$. Consequently, we have an embedding

$$\mathfrak{Y}_0 \to \mathfrak{g} \times \mathfrak{n} \times \mathfrak{g}, (a, b) \mapsto (a, [a, b], b).$$

Consider $\mathfrak{Y} = G \times^B \mathfrak{Y}_0$ fitting into the Cartesian diagram below and the identification

$$G \times^B (\mathfrak{g} \times \mathfrak{n} \times \mathfrak{g}) \cong T^*(\mathcal{B} \times \mathfrak{g}), (g, x, y, z) \mapsto (g, g \cdot x, g \cdot y, g \cdot z).$$

Let $i_{\mathfrak{Y}} : \mathfrak{Y} \to T^*(\mathcal{B} \times \mathfrak{g})$ be the composition of the bottom two maps in the diagram.

$$\begin{array}{ccccc} \mathfrak{Y}_0 & \longrightarrow & \mathfrak{g} \times \mathfrak{n} \times \mathfrak{g} & & \\ \downarrow & & \downarrow & & \\ G \times^B \mathfrak{Y}_0 & \longrightarrow & G \times^B (\mathfrak{g} \times \mathfrak{n} \times \mathfrak{g}) & \xrightarrow{\sim} & T^*(\mathcal{B} \times \mathfrak{g}) \end{array}$$

For any weight λ, let $\mathcal{L}_\lambda$ be the $\overline{G}$-equivariant line bundle on $\mathcal{B}$ associated to λ and $\widetilde{\mathcal{L}}_\lambda$ be the pullback of $\mathcal{L}_\lambda$ to $T^*(\mathcal{B} \times \mathfrak{g})$. Endow $(i_{\mathfrak{Y}})_* \mathcal{O}_{\mathfrak{Y}}$, resp. $(i_{\mathfrak{Y}})_* \mathcal{O}_{\mathfrak{Y}} \boxtimes \mathcal{O}_V$, with the trivial $\overline{G}$-equivariant structure.

We integrate the $\overline{\mathfrak{g}}$-action on $\widetilde{\mathrm{gr}}\ \mathbf{M}_c$, $\widetilde{\mathrm{gr}}\ \overline{\mathbf{M}}_c$, $\widetilde{\mathrm{gr}}\ \mathbf{N}_c$ and $\widetilde{\mathrm{gr}}\ \overline{\mathbf{N}}_c$ via the embedding τ_{-c} as in Construction 3.1. In this way, we have the following descriptions:

Corollary 3.6. *There are $\overline{G}$-equivariant isomorphisms:*

- $\widetilde{\mathrm{gr}}^{\mathrm{H}} \mathbf{M}_c \cong \widetilde{\mathcal{L}}_{\lceil \mu_c \rceil + \alpha} \otimes (i_{\mathfrak{Y}})_* \mathcal{O}_{\mathfrak{Y}}$;
- $\widetilde{\mathrm{gr}}^{\mathrm{H}} \overline{\mathbf{M}}_c \cong \left(\widetilde{\mathcal{L}}_{\lceil \mu_c \rceil + \alpha} \otimes (i_{\mathfrak{Y}})_* \mathcal{O}_{\mathfrak{Y}} \right) \boxtimes \mathcal{O}_V$.

Proof. We argue for $\widetilde{\mathrm{gr}}^{\mathrm{H}} \mathbf{M}_c$ only. By Proposition 3.3 and Lemma 3.5, we have

$$\widetilde{\mathrm{gr}}\ \mathbf{M}_c \cong \mathrm{ind}_B^G \widetilde{\mathrm{gr}}\ \mathbf{L}_c. \tag{34}$$

Therefore, by Lemma 3.2, the isomorphism in the corollary holds on a G-equivariant level. Moreover, by definition of the $\overline{G}$-action on $\widetilde{\mathrm{gr}}^{\mathrm{H}} \mathbf{M}_c$, the restriction of it to $\{\mathfrak{b}\} \times \mathfrak{g}$ is isomorphic to $\widetilde{\mathrm{gr}}\ \mathbf{L}$ as $\overline{B}$-equivariant sheaves. As a result, the isomorphism (34) holds $\overline{G}$-equivariantly. □

3.6. **Cuspidal DG modules.** We compute gr $\mathbf{N}_c$ closely following [Gin12, Section 4].

By the discussion in Section 3.5, the following diagram is Cartesian:

$$\begin{array}{ccc} \mathfrak{Y} & \xrightarrow{q_{\mathfrak{n}}} & G\times^B \mathfrak{n} \\ \downarrow & & \downarrow \\ G\times^B(\mathfrak{n}\times\mathfrak{m}) & \xrightarrow{q_{\mathfrak{b}}} & G\times^B \mathfrak{b} \end{array} \tag{35}$$

with

$$q_{\mathfrak{b}} : G\times^B(\mathfrak{n}\times\mathfrak{m}) \to G\times^B \mathfrak{b}, \quad (g,x,y)\mapsto (g,[x,y])$$

and $q_{\mathfrak{n}}$ is the restriction of $q_{\mathfrak{b}}$ to $\mathfrak{Y}$.

On $\mathcal{B}$ we have the vector bundle $\underline{\mathfrak{b}^*}$ (resp. $\underline{\mathfrak{b}}$) whose total space equals $G\times^B \mathfrak{b}^*$ (resp. $G\times^B \mathfrak{b}$). Let $\pi_{\mathfrak{b}} : G\times^B \mathfrak{b}\to\mathcal{B}$ be the projection and $\iota_{\mathfrak{b}} : \mathcal{B}\to G\times^B\mathfrak{b}$ be the zero section. The Koszul complex $(\wedge^\bullet \pi_{\mathfrak{b}}^*\underline{\mathfrak{b}^*}, \partial_{\mathfrak{b}})$, with differential $\partial_{\mathfrak{b}}$ defined by contraction with the canonical section of $\pi_{\mathfrak{b}}^*\underline{\mathfrak{b}}$, is quasi-isomorphic to $(\iota_{\mathfrak{b}})_*\mathcal{O}_{\mathcal{B}}$.

One similarly defines $\pi_{\mathfrak{n}}$, $\iota_{\mathfrak{n}}$, $\underline{\mathfrak{n}^*}$, $\partial_{\mathfrak{n}}$, such that $(\wedge^\bullet\pi_{\mathfrak{n}}^*\underline{\mathfrak{n}^*},\partial_{\mathfrak{n}})$ is quasi-isomorphic to $(\iota_{\mathfrak{n}})_*\mathcal{O}_{\mathcal{B}}$.

We define a DG algebra by

$$\mathscr{A} := ((\wedge^\bullet(\pi_{\mathfrak{n}}\circ q_{\mathfrak{n}})^*\underline{\mathfrak{n}^*}, q_{\mathfrak{n}}^*\partial_{\mathfrak{n}})).$$

By definition, the associated DG scheme $Spec(\mathscr{A})$ over $\mathfrak{Y}$ makes the following diagram Cartesian.

$$\begin{array}{ccc} Spec(\mathscr{A}) & \longrightarrow & \mathcal{B} \\ \downarrow & & \downarrow{\scriptstyle \iota_{\mathfrak{n}}} \\ \mathfrak{Y} & \xrightarrow{q_{\mathfrak{n}}} & G\times^B\mathfrak{n} \end{array} \tag{36}$$

Let $i_{\mathfrak{Y}\to G\times^B(\mathfrak{n}\times\mathfrak{m})}$ denote the embedding indicated by its subscript. Given the Cartesian diagram (35), we have that $(i_{\mathfrak{Y}\to G\times^B(\mathfrak{n}\times\mathfrak{m})})_*\mathscr{A}$ is quasi-isomorphic to

$$\mathcal{A} := (\wedge^\bullet(\pi_{\mathfrak{b}}\circ q_{\mathfrak{b}})^*\underline{\mathfrak{b}^*}, q_{\mathfrak{b}}^*\partial_{\mathfrak{b}}).$$

Because (36) and the diagram on the left of (37) below are Cartesian, the diagram on the right of (37) is also Cartesian.

$$\begin{array}{ccc} G\times^B T^*\mathfrak{g} & \longrightarrow & \mathcal{B} \\ \downarrow & & \downarrow \\ T^*(\mathcal{B}\times\mathfrak{g}) & \longrightarrow & G\times^B\mathfrak{n} \end{array} \qquad \begin{array}{ccc} Spec(\mathscr{A}) & \longrightarrow & \mathcal{B}\times T^*\mathfrak{g} \\ \downarrow & & \downarrow \\ \mathfrak{Y} & \longrightarrow & T^*(\mathcal{B}\times\mathfrak{g}) \end{array} \tag{37}$$

Diagrams (35), (36) and (37) can be combined into Figure 1.

Below, we denote the projection from a variety X to $\mathcal{B}$ by π_X.

Definition 3.3. The cuspidal DG module of slope c is

$$\mathcal{A}_c := \mathcal{A}\otimes\pi_{G\times^B(\mathfrak{n}\times\mathfrak{m})}^*\mathcal{L}_{\lceil\mu_c\rceil} \tag{38}$$

where $\lceil\mu_c\rceil$ is as defined in (29).

Similarly, define $\mathscr{A}_c := \mathscr{A}\otimes\pi_{\mathfrak{Y}}^*\mathcal{L}_{\lceil\mu_c\rceil}$.

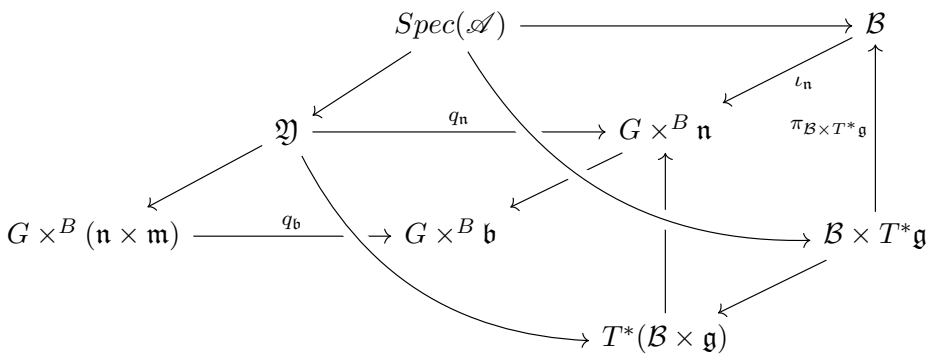
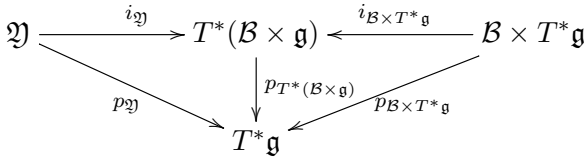


FIGURE 1. Cartesian diagrams

Consider the following maps, where $p_{\mathfrak{Y}} : \mathfrak{Y} \to T^*\mathfrak{g}$ is the restriction of $p : G \times^B (\mathfrak{n} \times \mathfrak{m}) \to T^*\mathfrak{g} : (g, x, y) \mapsto (g \cdot x, g \cdot y)$ and other maps are either natural projections or embeddings.



Proposition 3.7. *There are $\overline{G}$-equivariant isomorphisms:*

$$\widetilde{\mathrm{gr}}\ \mathbf{N}_c \cong Rp_*\mathcal{A}_c \qquad \widetilde{\mathrm{gr}}\ \overline{\mathbf{N}}_c \cong Rp_*\mathcal{A}_c \boxtimes \mathcal{O}_V.$$

Proof. We only argue for $\widetilde{\mathrm{gr}}\ \mathbf{N}_c$. By Theorem 3.4, we have a G-equivariant isomorphism:

$$\widetilde{\mathrm{gr}}\ \mathbf{N}_c \cong R(p_{\mathcal{B}\times T^*\mathfrak{g}})_* \left(Li^*_{\mathcal{B}\times T^*\mathfrak{g}}(\widetilde{\mathrm{gr}}\ \mathbf{M}_c) \otimes \pi^*_{\mathcal{B}\times T^*\mathfrak{g}}\omega_{\mathcal{B}} \right). \tag{39}$$

Moreover, by construction of the $\overline{G}$-equivariant structure on $\widetilde{\mathrm{gr}}\ \mathbf{N}_c$ and $\widetilde{\mathrm{gr}}\ \mathbf{M}_c$, the isomorphism (39) respects the $\overline{G}$-equivariance.

We plug the description of $\widetilde{\mathrm{gr}}\ \mathbf{M}_c$ from Corollary 3.6: into (39) and obtain

$$\widetilde{\mathrm{gr}}\ \mathbf{N}_c = R(p_{\mathcal{B}\times T^*\mathfrak{g}})_* \left(Li^*_{\mathcal{B}\times T^*\mathfrak{g}} \left(\widetilde{\mathcal{L}}_{\lceil \mu_c \rceil + \alpha} \otimes (i_{\mathfrak{Y}})_*\mathcal{O}_{\mathfrak{Y}} \right) \otimes \pi^*_{\mathcal{B}\times T^*\mathfrak{g}}\omega_{\mathcal{B}} \right). \tag{40}$$

The following is a simple consequence of base change (cf. [Gin12, Lemma 4.4.1]):
Suppose X is a smooth variety and $i_Y : Y \to X$, $i_Z : Z \to X$ are embeddings of closed subvarieties. Then:

$$(i_Y)_*Li^*_Y(i_Z)_*\mathcal{O}_Z \cong (i_Y)_*\mathcal{O}_Y \otimes_{\mathcal{O}_X} (i_Z)_*\mathcal{O}_Z \cong (i_Z)_*Li^*_Z(i_Y)_*\mathcal{O}_Y.$$

Applying this to (40) in the setting of $Y = \mathcal{B} \times T^*\mathfrak{g}$ and $Z = \mathfrak{Y}$, we obtain:

$$\begin{aligned}
(40) \cong & \mathrm{R}(p_{T^*(\mathcal{B}\times\mathfrak{g})})_* \left(\widetilde{\mathcal{L}}_{\lceil \mu_c \rceil} \otimes (i_{\mathfrak{Y}})_*\mathcal{O}_{\mathfrak{Y}} \otimes (i_{\mathcal{B}\times T^*\mathfrak{g}})_*\mathcal{O}_{\mathcal{B}\times T^*\mathfrak{g}} \right) \\
\cong & R(p_{T^*(\mathcal{B}\times\mathfrak{g})})_*R(i_{\mathfrak{Y}})_*L(i_{\mathfrak{Y}})^* \left(\widetilde{\mathcal{L}}_{\lceil \mu_c \rceil} \otimes (i_{\mathcal{B}\times T^*\mathfrak{g}})_*\mathcal{O}_{\mathcal{B}\times T^*\mathfrak{g}} \right) \\
\cong & R(p_{\mathcal{B}\times T^*\mathfrak{g}})_*L(i_{\mathfrak{Y}})^* \left(\widetilde{\mathcal{L}}_{\lceil \mu_c \rceil} \otimes (i_{\mathcal{B}\times T^*\mathfrak{g}})_*\mathcal{O}_{\mathcal{B}\times T^*\mathfrak{g}} \right). \qquad (41)
\end{aligned}$$

Since the diagram on the right of (37) is Cartesian, we have

$$R(i_{\mathfrak{Y}})_* \left(\widetilde{\mathcal{L}}_{\lceil \mu_c \rceil} \otimes (i_{\mathcal{B}\times T^*\mathfrak{g}})_*\mathcal{O}_{\mathcal{B}\times T^*\mathfrak{g}} \right) \cong R(i_{\mathfrak{Y}})_*\mathscr{A}_c.$$

Therefore, (41) equals

$$R(p_{\mathcal{B}\times T^*\mathfrak{g}})_*R(i_{\mathfrak{Y}})_*\mathscr{A}_c \cong R(p_{\mathfrak{Y}})_*\mathscr{A}_c \cong Rp_*\mathcal{A}_c. \qquad \square$$

4. K-theory of Hilbert schemes and bigraded characters

4.1. **BKR equivalence for** Hilb^n**.** Write $A := \mathbb{C}^* \times \mathbb{C}^*$.

The isospectral Hilbert scheme IHilb^n is the *reduced* fibered product of the following diagram.

$$\begin{array}{ccc} \mathrm{IHilb}^n & \longrightarrow & \mathbb{C}^{2n} \\ \downarrow{\scriptstyle \alpha} & & \downarrow \\ \mathrm{Hilb}^n & \longrightarrow & \mathbb{C}^{2n}/S_n \end{array}$$

A deep result of Haiman [Hai01] states that $\mathcal{P} := \alpha_* \mathcal{O}_{\mathrm{IHilb}^n}$ is a vector bundle of rank $n!$, which is known as the Procesi bundle $\mathcal{P}$. As a corollary, one has the following special case of the Bridgeland-King-Reid equivalence between Grothendieck groups:

$$\Gamma(\mathrm{Hilb}^n, \mathcal{P} \otimes (-)) : K^A(\mathrm{Hilb}^n) \cong K^{S_n \times A}(\mathbb{C}^{2n}). \tag{42}$$

Define the algebra of symmetric polynomials in infinite many variables:

$$K := \mathbb{C}(q,t)[z_1, z_2, \dots]^{S_\infty}. \tag{43}$$

For every partition $\lambda \vdash n$, let V_λ be the associated irreducible representation of S_n. The Grothendieck group $K^{S_n \times A}(\mathbb{C}^{2n})$ is freely generated by $V_\lambda \otimes \mathbb{C}[\mathbb{C}^{2n}]$ over $\mathbb{C}[q^\pm, t^\pm]$. Let $s_\lambda \in K$ be the Schur function associated to $\lambda \vdash n$. The Frobenius character $\mathrm{ch}_{S_n} : \mathrm{Rep}(S_n) \to \mathbb{C}[z_1, z_2, \dots]^{S_\infty}$ is an additive map sending V_λ to s_λ. For a bigraded S_n-representation $M = \bigoplus_{i,j} M_{i,j}$, one may further consider its bigraded Frobenius character valued in K:

$$\mathrm{ch}_{S_n \times A}(M) = \sum_{i,j} q^i t^j \mathrm{ch}_{S_n}(M_{i,j}).$$

Write $Z = z_1 + \cdots + z_n$. For every $f \in K$, let $f[Z]$ the plethystic substitution of Z into f. The bigraded Frobenius character of $V_\lambda \otimes \mathbb{C}[\mathbb{C}^{2n}]$ is (cf. [Hai03, Proposition 3.3.1])

$$\mathrm{ch}_{S_n \times A}(V_\lambda \otimes \mathbb{C}[\mathbb{C}^{2n}]) = s_\lambda \left[\frac{Z}{(1-q)(1-t)} \right].$$

Therefore, composing (42) with $\mathrm{ch}_{S_n \times A}$ and taking a direct sum establishes an isomorphism:

$$\kappa : \bigoplus_n K^A(\mathrm{Hilb}^n) \cong \{ f \in K | f\,[(1-q)(1-t)Z] \text{ has coefficients in } \mathbb{C}[q^\pm, t^\pm] \}, \tag{44}$$

such that $\kappa(\mathcal{V}_\lambda) = s_\lambda \left[\frac{Z}{(1-q)(1-t)} \right]$ where $\mathcal{V}_\lambda = \mathrm{Hom}_{S_n}(V_\lambda, \mathcal{P})$.

After localization, we have an identification:

$$\bigoplus_n \mathbb{C}(q,t) \otimes_{\mathbb{C}[q^\pm, t^\pm]} K^A(\mathrm{Hilb}^n) \cong K. \tag{45}$$

4.2. **Macdonald polynomials and fixed points.** For $\lambda \vdash n$, let p_λ denote the associated power sum symmetric polynomial. On K, we have an inner product [Mac95]:

$$(p_\lambda, p_\mu) = \delta_{\lambda\mu} z_\lambda \prod_i \frac{1-q^{\lambda_i}}{1-t^{\lambda_i}}, \quad \text{where } z_{(1^{n_1} 2^{n_2} \dots)} = \prod_i i^{n_i} n_i!$$

Let λ^t denote the transpose of the partition $\lambda \vdash n$. The modified Macdonald polynomials $\widetilde{H}_\lambda(z;q,t)$ are the unique q,t-symmetric polynomials satisfying

$$\begin{aligned}
&\widetilde{H}_\lambda\left[(1-q)z;q,t\right] \in \mathbb{C}(q,t)\langle s_\mu | \mu \geq \lambda\rangle;\\
&\widetilde{H}_\lambda\left[(1-t)z;q,t\right] \in \mathbb{C}(q,t)\langle s_\mu | \mu \geq \lambda^t\rangle;\\
&\widetilde{H}_\lambda[1;q,t] = (\widetilde{H}_\lambda, s_{(n)}) = 1.
\end{aligned} \tag{46}$$

The A-fixed points in Hilb^n are monomial ideals of co-length n, which are in bijection with partitions of n. For any $\lambda \vdash n$, let I_λ be the associated fixed point and $[I_\lambda]$ be the K-theory class corresponding to the skyscraper sheaf supported on I_λ.

Proposition 4.1. *(*[Hai03, Theorem 4.1.5, Proposition 5.4.1]*) The image of* $[I_\lambda]$ *under (44) is* $\widetilde{H}_\lambda$.

4.3. **Localization formula.** Throughout, we do not distinguish between partitions and Young diagrams. For a box x inside a Young diagram $\lambda \vdash n$, let a, ℓ, resp. a', ℓ', denote its arm and leg, resp. coarm and coleg, demonstrated below.

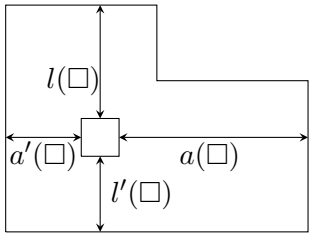


FIGURE 2.

For a representation of V of A with weight space decomposition $V = \bigoplus_{i,j} V_{i,j}$, we write

$$\mathrm{ch}_A(V) = \sum q^i t^j \dim(V_{i,j}).$$

and

$$\Lambda(V) = \sum_{i=0}^{\dim(V)} (-1)^i \mathrm{ch}_A(\wedge^i V).$$

By [ESm88, Hai98],

$$\Lambda(T^*_{I_\lambda}\mathrm{Hilb}^n) = \prod_{x\in\lambda}(1-q^{a(x)+1}t^{-\ell(x)})(1-q^{-a(x)}t^{\ell(x)+1}) =: g_\lambda. \tag{47}$$

For any $[\mathcal{F}] \in K^A(\mathrm{Hilb}^n)$, under the isomorphism (44), we have the localization formula [CG10, Remark 5.11.8]:

$$\kappa([\mathcal{F}]) = \sum_{\lambda\vdash n} \frac{\widetilde{H}_\lambda}{g_\lambda} \mathrm{ch}_A(\mathcal{F}|_{I_\lambda}). \tag{48}$$

The notation $\mathcal{F}|_{I_\lambda}$ denotes the **derived** pullback of $\mathcal{F}$ along the embedding $\{I_\lambda\} \hookrightarrow X$. **This convention will be adopted henceforth.**

To compute $\mathrm{ch}_{S_n\times A}(\mathrm{L}_c)$, we will apply the localization formula to $\Psi_c(\overline{\mathbf{N}}_c)$. For this goal, we will compute $\mathrm{ch}_A\left(\Psi_c(\overline{\mathbf{N}}_c)|_{I_c}\right)$ next.

4.4. **Principal nilpotent pairs.** We recollect some results of [Gin00, Gin12], which are valid for any semisimple Lie group G with Lie algebra $\mathfrak{g}$.

Let $\mathfrak{C} := \{(x,y) \in \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}, [x,y] = 0\}_{\mathrm{red}}$ be the commuting variety.

For any G-variety X, we write X_r to denote the regular locus, i.e. when the stablizer of the G-action is of the minimal possible dimension. The subvariety $\mathfrak{C}_r \subset \mathfrak{C}$ is smooth and open dense.

Definition 4.1. ([Gin00]) An element $\mathbf{e} = (x_1, x_2)$ in $\mathfrak{C}_r$ is called a principal nilpotent pair if for any $(t_1, t_2) \in \mathbb{C}^* \times \mathbb{C}^*$, there exists some $g \in G$ such that $(t_1 x_1, t_2 x_2) = (\mathrm{Ad}(g)x_1, \mathrm{Ad}(g)x_2)$.

It is shown in [Gin00, Theorem 1.2] that for every principal nilpotent pair (e_1, e_2), there exists an associated semisimple pair $\mathbf{h} = (h_1, h_2)$ such that $\mathbf{h}$ is regular and

$$[h_i, e_j] = \delta_{ij} e_j, \quad i, j = 1, 2.$$

The adjoint action of (h_1, h_2) decomposes $\mathfrak{g}$ into weight spaces

$$\mathfrak{g} = \bigoplus_{a,b \in \mathbb{Z}^2} \mathfrak{g}_{a,b}, \quad \text{where } \mathrm{ad}(h_1)x = ax, \mathrm{ad}(h_2)x = bx, \forall x \in \mathfrak{g}_{a,b}.$$

For every fixed principal nilpotent pair $\mathbf{e}$ with an associated semisimple pair $\mathbf{h}$, let $\rho : A \to G$ be the 2-parameter subgroup with differential at the identity being $\mathbb{C}^2 \to \mathfrak{g}$: $(1,0) \mapsto h_1$, $(0,1) \mapsto h_2$. We define an A-action on $\mathfrak{g}$ by $\mathrm{Ad}(\rho)$ and an A-action $\mathfrak{g} \times \mathfrak{g}$ by

$$(t_1, t_2)(x, y) = (t_1^{-1}\mathrm{Ad}(\rho(t_1, t_2))x, t_2^{-1}\mathrm{Ad}(\rho(t_1, t_2))y) \tag{49}$$

such that $\mathbf{e}$ is a fixed point under this action. The weight space $(\mathfrak{g} \oplus \mathfrak{g})_{a,b}$ under the action (49) equals $\mathfrak{g}_{a+1,b} \oplus \mathfrak{g}_{a,b+1}$.

4.5. **Principal nilpotent pairs for $\mathfrak{sl}_n$.** In the case of $\mathfrak{sl}_n$, conjugacy classes of principal nilpotent pairs are in bijection with A-fixed points in Hilb^n, or partitions of n. Indeed, for a partition $\lambda \vdash n$, let e_1, resp. e_2 be the associated Jordan normal form associated to λ, resp. λ^t. Then $\mathbf{e}_\lambda := (e_1, e_2)$ defines a principal nilpotent pair and all principal nilpotent pairs up to conjugation can be constructed in this way.

Definition 4.2. ([Neg15b, 4.5]) A Young tableau is called **almost** standard if the labels increase rightwards on rows and upwards on columns, with the exception that the labels are allowed to decrease up to 1 going up.

For an example, see Appendix B.2.2 for a full list of all almost standard Young tableaux (ASYT) of three boxes. Recall that if the labels increase rightwards on rows and upwards on columns, we obtain a standard Young tableau. Let ASYT_λ, resp. SYT_λ, be the set of almost standard Young tableaux, resp. standard Young tableaux, of shape λ.

Remark 4.1. Almost standard Young tableaux first appear in the discussion of the "eccentric correspondence" in [Neg15b, 4.5] and [GN24, 2.3], which captures the geometry of the cuspidal DG algebra $\mathcal{A}$ at homological degree 0 when specialized.

Every $\sigma \in \mathrm{SYT}_\lambda$ gives a semisimple pair associated to $\mathbf{e}_\lambda$ [Gin00, Section 5]: suppose for every label $k \in [1, n]$ the corresponding box $\square_k$ is at position (a_k, b_k), i.e, $a'(\square_k) = a_k$ and $\ell'(\square_k) = b_k$ (cf. Figure 4.3). Define diagonal matrices

$$h_1 = (a_n, \ldots, a_1), \quad h_2 = (b_n, \ldots, b_1). \tag{50}$$

The pair (h_1, h_2) satisfies the desired properties. For example, under the notations (26), when $(e_1, e_2) = (\underline{e}, 0)$, there is a unique associated semisimple pair given by $(h_1, h_2) = (\frac{1}{2}\underline{h}, 0)$.

Fix a principal nilpotent pair $\mathbf{e}_\lambda$ with an associated semisimple pair $\mathbf{h}_\lambda$. Since $\mathbf{h}_\lambda$ is regular, the Borel subalgebras containing both h_1 and h_2 are in bijection with the Weyl group. We fix such a

bijection $w \leftrightarrow \mathfrak{b}_w$. Recall the space $\mathfrak{Y} = G \times^B (\mathfrak{n} \times \mathfrak{m})$ from (35). For every $\mathfrak{b}_w$, we define associated $\mathfrak{n}_w$ and $\mathfrak{m}_w$. According to [BG13, Lemma 4.4.1], there is a bijection

$$\{\mathfrak{b}_w, w \in W | e_1 \in \mathfrak{n}_w, e_2 \in \mathfrak{n}_w\} \leftrightarrow \mathrm{SYT}_\lambda.$$

Similarly,

Lemma 4.2. *There is a bijection:*

$$\{\mathfrak{b}_w, w \in W | e_1 \in \mathfrak{n}_w, e_2 \in \mathfrak{m}_w\} \leftrightarrow \mathrm{ASYT}_\lambda.$$

We will write $w \in \mathrm{ASYT}_\lambda$ when $\mathfrak{b}_w$ satisfies the condition in Lemma 4.2.

Consider the fixed point set under the A-action defined by (49):

$$\mathfrak{Y}_r^A = \sqcup_{w \in W} \mathfrak{Y}_r^{w,A} \quad \text{with } \mathfrak{Y}_r^{w,A} := \{\mathfrak{b}_w\} \times (\mathfrak{n}_w \oplus \mathfrak{m}_w) \cap (\mathfrak{g} \oplus \mathfrak{g})_r^A.$$

Though $[\mathfrak{n}_w, \mathfrak{m}_w] = \mathfrak{b}_w$, we will insist on writing $[\mathfrak{n}_w, \mathfrak{m}_w]$ to emphasize the A-action on it is induced by composing (49) with $[-,-]$. Fix an A-stable subspace $\mathfrak{R}_w \subset [\mathfrak{n}_w, \mathfrak{m}_w]$ such that

$$[\mathfrak{n}_w, \mathfrak{m}_w] = \mathfrak{R}_w \oplus [\mathfrak{n}_w, \mathfrak{m}_w]^A.$$

4.6. **Bigraded characters.** By Lemma 2.3, $\widetilde{\mathrm{gr}}\ \mathbf{N}_c$ is supported on $\mathfrak{C}$. Let $\mathscr{F}_c$ be the (nonderived) restricdtion of $\widetilde{\mathrm{gr}}\ \mathbf{N}_c$ to $\mathfrak{C}_r$.

The following lemma mimics [BG13, Proposition 3.8.6] and the proof follows verbatim, via replacing their variety $G \times^B (\mathfrak{b} \times \mathfrak{b})$ by our $\mathfrak{Y}$.

Lemma 4.3. $\mathrm{ch}_A(\mathscr{F}_c|_{\mathbf{e}_\lambda}) = \Lambda\left((T^*_{\mathfrak{C}_r^A}\mathfrak{C}_r)|_{\mathbf{e}_\lambda}\right) \sum_{w \in \mathrm{ASYT}_\lambda} \Lambda\left((T^*_{\mathfrak{Y}_r^{w,A}}\mathfrak{Y}_r)|_{\mathbf{e}_\lambda}\right)^{-1} \Lambda\left(\mathfrak{R}_w^* \otimes \mathbb{C}_{\lceil \mu_c \rceil}\right).$

We adopt the Ω-notation by setting

$$\Omega(\sum_{i,j} a_{i,j} q^i t^j) = \prod_{i,j} (1 - q^i t^j)^{a_{i,j}}, \quad \Omega^0(F) = \Omega(F - a_{0,0}) \tag{51}$$

such that

$$\Lambda(V) = \Omega(\mathrm{ch}_A(V)), \quad \Lambda(V/V^A) = \Omega^0(\mathrm{ch}_A(V)).$$

For a Young tableau, define the weight of the box x labeled i by

$$\chi_i = q^{a'(x)} t^{l'(x)}.$$

We also write

$$\omega(x) = \frac{(1-x)(1-qtx)}{(1-qx)(1-tx)} = \Omega\left((1-q)(1-t)x\right). \tag{52}$$

The following proposition is an analogue to [BG13, Theorem 4.5.1].

Proposition 4.4. *The following identity holds*

$$\mathrm{ch}_A(\mathscr{F}_c|_{\mathbf{e}_\lambda}) = g_\lambda \frac{(1-qt)^{n-1}}{(1-t)^{n-1}(-t)^{n-1}} \sum_{\sigma \in \mathrm{ASYT}_\lambda} \frac{\Xi_\sigma \prod_{i=1}^n \chi_{n-i+1}^{\lfloor \mu_c \rfloor(i)}}{\hat{\prod}_{i=1}^{n-1}(1 - \frac{\chi_i}{t\chi_{i+1}})} \tag{53}$$

where

$$\Xi_\sigma := \hat{\prod}_i \frac{1}{(1-\chi_i^{-1})} \hat{\prod}_{1 \le i < j \le n} \omega(\frac{\chi_i}{\chi_j}). \tag{54}$$

The "restricted" product $\hat{\prod}$ means we drop all the zero linear denominators.

Proof. We compute every term on the right hand side of Lemma 4.3 explicitly.

First of all, in view of (49) and (50), we have that

$$\mathrm{ch}_A(\mathbb{C}_{\lceil \mu_c \rceil}) = \prod_{i=1}^{n} \chi_{n-i+1}^{\lceil \mu_c \rceil (i)}.$$

Below, to ease notations, we will simply write $\mathbf{e} := \mathbf{e}_\lambda$ and $\mathrm{Stab}(\mathbf{e}) := \mathrm{Stab}_{\mathfrak{g}}(\mathbf{e})$.

By definition, $\Lambda(\mathfrak{R}_w^*) = \Lambda(([\mathfrak{n}_w, \mathfrak{m}_w]/[\mathfrak{n}_w, \mathfrak{m}_w]^A)^*)$. Moreover, by [BG13, Lemma 3.9.1],

$$\Lambda\left((T^*_{\mathfrak{C}_r^A}\mathfrak{C}_r)|_{\mathbf{e}}\right) = g_\lambda \cdot \Lambda\left((\mathfrak{g}/(\mathrm{Stab}(\mathbf{e}) \oplus \mathfrak{h}))^*\right);$$
$$\Lambda\left((T^*_{\mathfrak{Y}_r^{w,A}}\mathfrak{Y}_r)|_{\mathbf{e}}\right) = \Lambda(\mathfrak{n}_w) \cdot \Lambda\left(\left((\mathfrak{n}_w \oplus \mathfrak{m}_w)/(\mathfrak{g} \oplus \mathfrak{g})^A \cap (\mathfrak{n}_w \oplus \mathfrak{m}_w)\right)^*\right).$$

Let R, resp. R^+, denote the set of all roots, resp. positive roots with respect to $\mathfrak{b}_w$ and $\Delta \subset R^+$ be the set of simple roots. Write

$$\overline{\Sigma} := \sum_{1 \le i < j \le n} \chi_i \chi_j^{-1}, \quad \underline{\Sigma} := \sum_{1 \le i < j \le n} \chi_i^{-1} \chi_j.$$

and recall that $\Lambda(V) = \Omega(\mathrm{ch}_A(V))$ and $\Lambda(V/V^A) = \Omega^0(\mathrm{ch}_A(V))$.

We have identities:

$$\mathrm{ch}_A(\mathfrak{h}^*) = n - 1$$
$$\mathrm{ch}_A(\mathfrak{g}^*) = n - 1 + \sum_{\alpha_1, \alpha_2 \in R} q^{\alpha_1(h_1)} t^{\alpha_2(h_2)} = n - 1 + \overline{\Sigma} + \underline{\Sigma}$$
$$\mathrm{ch}_A(\mathrm{Stab}(\mathbf{e})^*) = \sum_{(a,b) \in \lambda} q^{-a} t^{-b} = \sum_{i=1}^{n} \chi_i^{-1}, \qquad (\because \mathrm{Stab}(\mathbf{e}) = \mathbb{C}[e_1, e_2])$$

which imply that

$$\Lambda\left([\mathfrak{g}/(\mathrm{Stab}(\mathbf{e}) \oplus \mathfrak{h})]^*\right) = \Omega(\overline{\Sigma} + \underline{\Sigma})\Omega(\sum_{i=1}^{n} \chi_i^{-1})^{-1}.$$

The identity

$$\mathrm{ch}_A(\mathfrak{n}_w) = \sum_{\alpha \in R^+} q^{\alpha(h_1)} t^{\alpha(h_2)} = \underline{\Sigma}$$

implies that $\Lambda(\mathfrak{n}_w) = \Omega(\underline{\Sigma})$. The identities

$$\mathrm{ch}_A(\mathfrak{n}_w^* \oplus \{0\}) = \sum_{\alpha \in R^+} q^{1-\alpha_1(h_1)} t^{-\alpha_2(h_2)} = q\overline{\Sigma}$$
$$\mathrm{ch}_A(\{0\} \oplus \mathfrak{m}_w^*) = \sum_{\alpha \in R^+} q^{-\alpha(h_1)} t^{1-\alpha(h_2)} + (n-1)t + \sum_{\alpha \in \Delta} q^{\alpha(h_1)} t^{1+\alpha(h_2)}$$
$$= (n-1)t + t\overline{\Sigma} + t\sum_{i=1}^{n-1} \chi_{i+1}\chi_i^{-1}$$

imply

$$\Lambda\left(\left[(\mathfrak{n}_w \oplus \mathfrak{m}_w)/(\mathfrak{g} \oplus \mathfrak{g})^A \cap (\mathfrak{n}_w \oplus \mathfrak{m}_w)\right]^*\right) = \Omega^0\left((n-1)t + (q+t)\overline{\Sigma} + t\sum_{i=1}^{n-1} \chi_{i+1}\chi_i^{-1}\right).$$

Finally,

$$\begin{aligned}\mathrm{ch}_A([\mathfrak{n}_w,\mathfrak{m}_w]) =&(n-1)qt+\sum_{\alpha\in R^+} q^{1+\alpha_1(h_1)}t^{1+\alpha_2(h_2)}\\ =&(n-1)qt+qt\overline{\Sigma}.\end{aligned}$$

implies

$$\Lambda(([\mathfrak{n}_w,\mathfrak{m}_w]/[\mathfrak{n}_w,\mathfrak{m}_w]^A)^*)=\Omega^0((n-1)qt+qt\overline{\Sigma}).$$

Combining the identities above, we deduce that:

$$\begin{aligned}&\frac{1}{g_\lambda}\Lambda((T^*_{\mathfrak{C}_r^A}\mathfrak{C}_r)|_{\mathbf{e}})\cdot\Lambda\left((T^*_{\mathfrak{Y}_r^{w,A}}\mathfrak{Y}_r)|_{\mathbf{e}}\right)^{-1}\cdot\Lambda(\mathfrak{R}_w^*)\\ =&\Lambda\left([\mathfrak{g}/(\mathrm{Stab}(\mathbf{e})\oplus\mathfrak{h})]^*\right)\cdot\left(\Lambda(\mathfrak{n}_w)\Lambda\left(\left[(\mathfrak{n}_w\oplus\mathfrak{m}_w)/(\mathfrak{g}\oplus\mathfrak{g})^A\cap(\mathfrak{n}_w\oplus\mathfrak{m}_w)\right]^*\right)\right)^{-1}\cdot\Lambda\left(([\mathfrak{n}_w,\mathfrak{m}_w]/\mathfrak{g}_{1,1})^*\right)\\ =&\left[\Omega(\overline{\Sigma}+\underline{\Sigma})\Omega(\sum_{i=1}^n\chi_i^{-1})^{-1}\right]\cdot\left[\Omega(\underline{\Sigma})\Omega^0((n-1)t+(q+t)\overline{\Sigma}+t\sum_{i=1}^{n-1}\chi_{i+1}\chi_i^{-1})\right]^{-1}\cdot\Omega^0\left((n-1)qt+qt\overline{\Sigma}\right)\\ =&\left(\frac{1-qt}{1-t}\right)^{n-1}\Omega^0((1+qt-q-t)\overline{\Sigma})\Omega(\sum_{i=1}^n\chi_i^{-1})^{-1}\Omega^0(t\sum_{i=1}^{n-1}\chi_{i+1}\chi_i^{-1})^{-1}\\ =&\left(\frac{1-qt}{1-t}\right)^{n-1}\sum_{\sigma\in\mathrm{ASYT}_{\mathbf{e}}}\frac{\hat{\prod}_{1\le i<j\le n}\omega(\frac{\chi_i}{\chi_j})}{\hat{\prod}_i(1-\chi_i^{-1})\hat{\prod}_{i=1}^{n-1}(1-t\frac{\chi_{i+1}}{\chi_i})}\\ =&\frac{(1-qt)^{n-1}}{(1-t)^{n-1}(-t)^{n-1}}\frac{\chi_1}{\chi_n}\sum_{\sigma\in\mathrm{ASYT}_{\mathbf{e}}}\frac{\hat{\prod}_{1\le i<j\le n}\omega(\frac{\chi_i}{\chi_j})}{\hat{\prod}_i(1-\chi_i^{-1})\hat{\prod}_{i=1}^{n-1}(1-\frac{\chi_i}{t\chi_{i+1}})}\end{aligned}$$

Now the equality

$$(\lceil\mu_c\rceil(1),\cdots,\lceil\mu_c\rceil(n))=(\lfloor\mu_c\rfloor(1),\cdots,\lfloor\mu_c\rfloor(n))+(1,0,\cdots,0,-1)$$

implies that

$$\frac{\chi_1}{\chi_n}\prod_{i=1}^n\chi_{n-i+1}^{\lceil\mu_c\rceil(i)}=\prod_{i=1}^n\chi_{n-i+1}^{\lfloor\mu_c\rfloor}$$

and the proposition is proved. □

Remark 4.2. On $\mathfrak{C}$, G acts diagonally and there is a Chevalley-type isomorphism $\mathbb{C}[\mathfrak{C}]^G\cong\mathbb{C}[\mathfrak{h}\times\mathfrak{h}]^W$. Let $\mathfrak{IC}$ be the fibered product defined by the diagram below.

$$\begin{array}{ccc}\mathfrak{IC} & \xrightarrow{\alpha} & \mathfrak{C}\\ \downarrow & & \downarrow\\ \mathfrak{h}^2 & \longrightarrow & \mathfrak{h}^2/W\end{array}$$

We note that the isospectral commuting variety in [Gin12] is defined in terms of the normalization of $\mathfrak{C}$ but the variety $\mathfrak{IC}$ is sufficient for our purpose as we will only work with its regular locus.

Define $\mathcal{R}:=\alpha_*\mathcal{O}_{\mathfrak{IC}}$. In *loc. cit.*, the restriction $\mathcal{R}|_{\mathfrak{C}_r}$ is proved to be a vector bundle, with a W-action on each fiber making it a regular representation of W. In the case of $\mathfrak{g}=\mathfrak{sl}_n$, take $(x,y,v)\in\widetilde{\mathrm{Hilb}^n}$. Having a cyclic vector implies that $(x,y)\in\mathfrak{C}_r$. By [Gin12, Corollary 8.2.5],

$$\left(\mathcal{R}|_{\mathfrak{C}_r}\boxtimes\mathcal{O}_V\right)|_{\widetilde{\mathrm{Hilb}^n}}=\pi^*\mathcal{P}\quad\text{ where }\pi:\widetilde{\mathrm{Hilb}^n}\to\mathrm{Hilb}^n.$$

According to [BG13, Theorem 4.5.1 and Proposition 4.8.2],

$$\mathrm{ch}_A(\mathcal{R}|_{\mathbf{e}_\lambda}) = \frac{g_\lambda}{(1-q)^n(1-t)^n} \sum_{\sigma \in \mathrm{SYT}_\lambda} \Xi_\sigma, \tag{55}$$

which coincides with the formula for $\mathrm{ch}_A(\mathcal{P}|_{I_\lambda})$ derived by Garsia and Haiman [GH98].

4.7. **Catalan DG modules.** We define an analogue of the cuspidal DG module. On $\mathcal{B}$ we also have the vector bundle $\underline{[\mathfrak{n},\mathfrak{n}]^*}$ (resp. $\underline{[\mathfrak{n},\mathfrak{n}]}$) whose total space equals $G \times^B [\mathfrak{n},\mathfrak{n}]^*$ (resp. $G \times^B [\mathfrak{n},\mathfrak{n}]$). Let $\pi_{[\mathfrak{n},\mathfrak{n}]} : G \times^B [\mathfrak{n},\mathfrak{n}] \to \mathcal{B}$ be the projection and $\iota_{[\mathfrak{n},\mathfrak{n}]} : \mathcal{B} \to G \times^B [\mathfrak{n},\mathfrak{n}]$ be the zero section. The Koszul complex $(\wedge^\bullet \pi^*_{[\mathfrak{n},\mathfrak{n}]} \underline{[\mathfrak{n},\mathfrak{n}]^*}, \partial_{[\mathfrak{n},\mathfrak{n}]})$, with differential given by contraction with the canonical section of $\pi^*_{[\mathfrak{n},\mathfrak{n}]} \underline{[\mathfrak{n},\mathfrak{n}]}$, is quasi-isomorphic to $(\iota_{[\mathfrak{n},\mathfrak{n}]})_* \mathcal{O}_{\mathcal{B}}$. Let

$$q_{[\mathfrak{n},\mathfrak{n}]} : G \times^B (\mathfrak{n} \times \mathfrak{n}) \to G \times^B [\mathfrak{n},\mathfrak{n}], \quad (g,x,y) \mapsto (g,[x,y]).$$

We define

$$\mathcal{A}' := \wedge^\bullet \left((\pi_{[\mathfrak{n},\mathfrak{n}]} \circ q_{[\mathfrak{n},\mathfrak{n}]})^* \underline{[\mathfrak{n},\mathfrak{n}]^*}, q^*_{[\mathfrak{n},\mathfrak{n}]} \partial_{[\mathfrak{n},\mathfrak{n}]} \right), \tag{56}$$

which fits into the Cartisian diagram:

$$\begin{array}{ccc} Spec(\mathcal{A}') & \longrightarrow & \mathcal{B} \\ \downarrow & & \downarrow{\scriptstyle \iota_{[\mathfrak{n},\mathfrak{n}]}} \\ G \times^B (\mathfrak{n} \times \mathfrak{n}) & \xrightarrow{q_{[\mathfrak{n},\mathfrak{n}]}} & G \times^B [\mathfrak{n},\mathfrak{n}] \end{array} \tag{57}$$

Define the Catalan DG module at slope c by

$$\mathcal{A}'_c = \mathcal{A}' \otimes (\pi_{G \times^B (\mathfrak{n} \times \mathfrak{n})})^* \mathcal{L}_{\lfloor \mu_c \rfloor}$$

where $\lceil \mu_c \rceil$ is as defined in (30). Note that the definitions (38) and (56) use different line bundles.

Write $p' : G \times^B (\mathfrak{n} \times \mathfrak{n}) \to \mathfrak{g} \times \mathfrak{g} : (g,x,y) \mapsto (g \cdot x, g \cdot y)$. Via diagram tracing, we have that $Rp'_* \mathcal{A}'_c$ is quasi-isomorphic to a complex of sheaves supported on $\mathfrak{C}$, whose restriction to $\mathfrak{C}_r$ will be denoted by $\mathscr{F}'_c$. Similar to Proposition 4.4, we have the following:

Proposition 4.5. *The following identity holds:*

$$\mathrm{ch}_A(\mathscr{F}'_c|_{\mathbf{e}_\lambda}) = g_\lambda \sum_{\sigma \in \mathrm{SYT}_\lambda} \frac{\Xi_\sigma \prod_{i=1}^n \chi_{n-i+1}^{\lfloor \mu_c \rfloor(i)}}{\hat{\prod}_{i=1}^{n-1} (1 - qt\frac{\chi_i}{\chi_{i+1}})} \tag{58}$$

where Ξ_σ is as defined in (54).

For any A-equivariant sheaf $\mathcal{F}$, we use the notation $q^a t^b \mathcal{F}$ to shift the original A-action by the weight (a,b). We will prove in the next sections that the sheaves $q^{1-n}\mathscr{F}_c$ and $\mathscr{F}'_c$ correspond to the same classes in $K^A(\mathrm{Hilb}^n)$ using shuffle algebra techniques. We conjecture that:

Conjecture 4.6. *There exists a $\overline{G} \times \mathbb{C}^* \times \mathbb{C}^*$-equivariant isomorphism:*

$$q^{1-n} Rp_* \mathcal{A}_c \cong Rp'_* \mathcal{A}'_c.$$

One should note that a priori it is not clear whether $Rp'_* \mathcal{A}'_c$ is concentrated in one degree. In contrast, the sheaf $Rp_* \mathcal{A}_c$ is automatically concentrated in one degree as it is the associated graded of $\mathbf{N}_c$.

4.8. **Shuffle algebras.** Take $S = \bigoplus_{n\geq 0} S_k$ for $S_k := \mathbb{C}(q,t)(z_1,\cdots,z_n)^{S_n}$. We endow S with a $\mathbb{C}(q,t)$-algebra structure via the shuffle product

$$f(z_1,\cdots,z_k) * g(z_1,\cdots,z_\ell) = \frac{1}{k!\ell!}\mathrm{Sym}\big[f(z_1,\cdots,z_k)g(z_{k+1},\cdots,z_{k+\ell})\prod_{i=1}^{k}\prod_{j=k+1}^{k+\ell}\omega(\frac{z_i}{z_j})\big].$$

Here Sym denotes symmetrization under the $S_{k+\ell}$-action and ω is as in (52).

Definition 4.3. The shuffle algebra $\mathfrak{A}$ is the subalgebra of S consisting of rational functions in the form of

$$F(z_1,\cdots,z_k) = \frac{f(z_1,\cdots,z_k)\prod_{1\leq i<j\leq k}(z_i-z_j)^2}{\prod_{1\leq i\neq j\leq k}(z_i-qz_j)(z_i-tz_j)} \tag{59}$$

such that $f \in \mathbb{C}[q^{\pm},t^{\pm}][z_1,\cdots,z_k]^{S_k}$ satisfying the wheel conditions:

$$f(z_1,z_2,z_3,\dots) = 0 \text{ if } \{\frac{z_1}{z_2},\frac{z_2}{z_3},\frac{z_3}{z_1}\} = \{q,t,\frac{1}{qt}\}.$$

Following [Neg22], we define[1]

$$P_{m,n} = \mathrm{Sym}\left(\frac{\prod_{i=1}^{n} z_{n-i+1}^{\lfloor ic\rfloor-\lfloor(i-1)c\rfloor}}{\prod_{i=1}^{n-1}(1-qt\frac{z_i}{z_{i+1}})}\prod_{1\leq i<j\leq n}\omega(\frac{z_i}{z_j})\right) \tag{60}$$

According to [SV13, Theorem 10.1], there is an isomorphism between $\mathfrak{A}$ and the positive half of the elliptic Hall algebra. Thus by [BS12], $P_{m,n}$ for $n\geq 1$, $m\in\mathbb{Z}$ generate the shuffle algebra $\mathfrak{A}$.

By [Neg22, (2.34) and (2.35)], (60) equals

$$P_{m,n} = \left(\frac{(1-qt)}{(1-t)(-qt)}\right)^{n-1}\mathrm{Sym}\left(\frac{\prod_{i=1}^{n} z_{n-i+1}^{\lfloor ic\rfloor-\lfloor(i-1)c\rfloor}}{\prod_{i=1}^{n-1}(1-\frac{z_i}{tz_{i+1}})}\prod_{1\leq i<j\leq n}\omega(\frac{z_i}{z_j})\right). \tag{61}$$

Theorem 4.7. *(*[FT11], [SV13, Corollary 1.5]*) There is a faithful action of the algebra $\mathfrak{A}$ on the vector space K.*

Proposition 4.8. *(*[Neg15a], [GN15, (49)]*) Under the action in Theorem 4.7,*

$$P_{m,n}\cdot 1 = \left(\frac{(1-q)(1-t)}{(1-qt)}\right)^{n}\sum_{\lambda\vdash n}\frac{\widetilde{H}_\lambda}{g_\lambda}\sum_{\sigma\in\mathrm{SYT}_\lambda}\frac{\prod_{i=1}^{n}\chi_{n-i+1}^{\lfloor ic\rfloor-\lfloor(i-1)c\rfloor-1}}{\hat{\prod}_{i=1}^{n-1}(1-qt\frac{\chi_i}{\chi_{i+1}})}\Theta_\sigma \tag{62}$$

$$=\frac{(1-q)^n(1-t)}{(1-qt)(-qt)^{n-1}}\sum_{\lambda\vdash n}\frac{\widetilde{H}_\lambda}{g_\lambda}\sum_{\sigma\in\mathrm{ASYT}_\lambda}\frac{\prod_{i=1}^{n}\chi_{n-i+1}^{\lfloor ic\rfloor-\lfloor(i-1)c\rfloor-1}}{\hat{\prod}_{i=1}^{n-1}(1-\frac{\chi_i}{t\chi_{i+1}})}\Theta_\sigma \tag{63}$$

where

$$\Theta_\sigma := \prod_{i=1}^{n}(1-qt\chi_i)\hat{\prod}_{1\leq i<j\leq n}\omega^{-1}(\frac{\chi_j}{\chi_i}). \tag{64}$$

Proof. The first identity is [Neg15a, Proposition 5.5] in terms of the definition (60) of $P_{m,n}$. To illustrate how ASYT appears, we include a proof of the second equality mimicking *loc. cit.* but with respect to the definition (61) of $P_{m,n}$ instead.

[1]Note that the $P_{m,n}$ here and in [Neg22] is denoted by $\widetilde{P}_{m,n}$ in [GN15], as a certain modification of the $P_{m,n}$ in [Neg14].

Using [Neg15a, Theorem 4.7] and localization formula (48), one has $P_{m,n} \cdot 1$ equal to

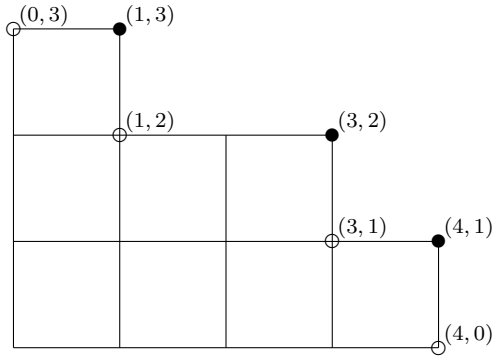
(65)
$$\left(\frac{(1-qt)}{(1-t)(-qt)}\right)^{n-1} \sum_{\lambda \vdash n} \frac{\widetilde{H}_\lambda}{g_\lambda} \int \frac{\prod_{i=1}^n z_{n-i+1}^{\lfloor ic \rfloor - \lfloor (i-1)c \rfloor}}{\prod_{i=1}^{n-1}(1-\frac{z_i}{tz_{i+1}})} \prod_{1\le i<j\le n} \omega(\frac{z_i}{z_j}) \prod_{\square \in \lambda} \prod_{i=1}^n \left(z_i^{-1} \omega^{-1}(\frac{z_i}{\chi(\square)})(1-qtz_i) \frac{dz_i}{2\pi i z_i} \right).$$

Each integral is taken along a contour separating poles of the function to be integrated.

For a partition $\lambda \vdash n$, [Neg15a, (5.4)] states (for a proof, cf. [BG13, Lemma 4.8.5])

$$\prod_{\square \in \lambda} \left(\omega^{-1}(\frac{z}{\chi(\square)})(1-qtz) \right) = \frac{\prod_{\circ \text{ inner corner of } \lambda}(1-\frac{qtz}{\chi(\circ)})}{\prod_{\bullet \text{ outer corner of } \lambda}(1-\frac{qtz}{\chi(\bullet)})}.$$

Notions of inner/outer corners are illustrated below. Hollow circles indicate the inner corners of the partition and solid circles indicate the outer corners. We use notations $\chi(\circ)$ and $\chi(\bullet)$ for weights of corners as well. For example, weights of outer corners in the figure below are qt^3, q^3t^2, q^4t.



[Neg15a, Figure 5.1]

Integrating over z_n results in a residue whenever qtz_n equals to the weight of some outer corner of the partition λ. Label the box adjacent to this corner by n. By change of variables, the residue is the weight of the box labeled by n. The integral in (65) thus becomes

$$\chi_n \int \left[\frac{\prod_{i=1}^n z_{n-i+1}^{\lfloor ic \rfloor - \lfloor (i-1)c \rfloor}}{\prod_{i=1}^{n-1}(1-\frac{z_i}{tz_{i+1}})} \hat{\prod}_{1\le i<j\le n} \omega(\frac{z_i}{z_j}) \hat{\prod}_{\square \in \lambda} \hat{\prod}_{i=1}^n \left(z_i^{-1} \omega^{-1}(\frac{z_i}{\chi(\square)})(1-qtz_i) \frac{dz_i}{2\pi i z_i} \right) \right] \Bigg|_{z_n = \chi_n}.$$

Notice the restricted products $\hat{\prod}$ here.

Next, when integrating over z_{n-1}, we obtain a residue whenever one of the following scenarios happens:

- qtz_{n-1} equals the weight of some outer corner of λ (i.e, z_{n-1} equals to the weight of some inner corner of λ).
- $q\frac{z_{n-1}}{\chi_n} = 1$ or $t\frac{z_{n-1}}{\chi_n} = 1$ (i.e, z_{n-1} equals to the weight of some box to the left or below the box labeled by n in the last step).
- $\frac{z_{n-1}}{t\chi_n} = 1$ (i.e, z_{n-1} equals to the weight of some box above the box labeled by n in the last step).

These three conditions together with the condition for z_n exactly define all the possible relative positions between the boxes labeled by n and $n-1$ in an ASYT such that the box labeled by n is in a corner. One can argue similarly starting from any label on a corner.

Repeating this procedure, we conclude that the integral in (65) equals

$$\sum_{\mathrm{ASYT}}\prod_{i=1}^{n}\chi_i\left[\frac{\prod_{i=1}^{n}z_{n-i+1}^{\lfloor ic\rfloor-\lfloor(i-1)c\rfloor}}{\hat{\prod}_{i=1}^{n-1}(1-\frac{z_i}{tz_{i+1}})}\hat{\prod}_{1\le i<j\le n}\omega(\frac{z_i}{z_j})\hat{\prod}_{\square\in\lambda}\hat{\prod}_{i=1}^{n}\left(\omega^{-1}(\frac{z_i}{\chi(\square)})(1-qtz_i)z_i^{-2}\right)\right]\Big|_{z_i=\chi_i,\forall i}$$

$$=\left(\frac{(1-q)(1-t)}{(1-qt)}\right)^n\sum_{\mathrm{ASYT}_\lambda}\left[\frac{\prod_{i=1}^{n}\chi_{n-i+1}^{\lfloor ic\rfloor-\lfloor(i-1)c\rfloor}}{\hat{\prod}_{i=1}^{n-1}(1-\frac{\chi_i}{t\chi_{i+1}})}\hat{\prod}_{1\le i<j\le n}\omega^{-1}(\frac{\chi_j}{\chi_i})\prod_{i=1}^{n}(1-qt\chi_i)\chi_i^{-1}\right].$$

The factor $\left(\frac{(1-q)(1-t)}{(1-qt)}\right)^n$ comes from $\hat{\prod}_{i=1}^{n}\omega^{-1}(\frac{\chi_i}{\chi_i})$. □

4.9. **"Cuspidal = Catalan" via shuffle generators.**

Proposition 4.9.

$$(P_{m+n,n}\cdot 1)=\sum_{\lambda\vdash n}\mathrm{ch}_A(\mathscr{F}'_{m/n}|_{\mathbf{e}_\lambda})\frac{\widetilde{H}_\lambda}{g_\lambda}$$

$$=q^{1-n}\sum_{\lambda\vdash n}\mathrm{ch}_A(\mathscr{F}_{m/n}|_{\mathbf{e}_\lambda})\frac{\widetilde{H}_\lambda}{g_\lambda}$$

Proof. Compring formulae (62)/(63) with (58)/(53), we see that it suffices to show that for every Young tableau σ with n boxes,

$$\frac{\Theta_\sigma}{\Xi_\sigma}=g_\sigma\frac{(1-qt)^n}{(1-q)^n(1-t)^n}. \tag{66}$$

By definition (54) and (64),

$$\frac{\Theta_\sigma}{\Xi_\sigma}=\Omega^0\left(qt\sum_i\chi_i+\sum_i\chi_i^{-1}-(1-q)(1-t)\sum_{1\le i<j\le n}\left(\frac{\chi_i}{\chi_j}+\frac{\chi_j}{\chi_i}\right)\right),$$

which only depends on the shape of σ. Same for the polynomial g_σ (47). Therefore, to show (66), we may assume that σ is a SYT.

For every positive integer $k\le n$, we let $\sigma(k)$ denote the Young sub-tableau of σ consisting of the first k labels. Then

$$\frac{\Theta_\sigma}{\Xi_\theta}=\prod_{j=1}^{n}\frac{\Theta_{\sigma(j)}}{\Xi_{\sigma(j)}}$$

with

$$\frac{\Theta_{\sigma(j)}}{\Xi_{\sigma(j)}}=\Omega^0\left(qt\chi_j+\chi_j^{-1}-(1-q)(1-t)\sum_{1\le i<j}\left(\frac{\chi_i}{\chi_j}+\frac{\chi_j}{\chi_i}\right)\right)$$

We let R_j (resp. C_j) denote the set of boxes of σ in the same row (resp. column) and to the left of (resp. below) the box labeled by j in the tableau $\sigma(k)$. For every box $x\in\sigma(k)$ we write $a_k(x),\ell_k(x)$ to indicate the arm and leg of x in $\sigma(k)$.

Then $g_\sigma=\prod_{j=1}^{n}\frac{g_{\sigma(j)}}{g_{\sigma(j-1)}}$ with $g_{\sigma(0)}=1$ and

$$\frac{g_{\sigma(j)}}{g_{\sigma(j-1)}}=(1-q)(1-t)\prod_{x\in C_j\cup R_j}\frac{1-q^{1+a_j(x)}t^{-\ell_j(x)}}{1-q^{1+a_{j-1}(x)}t^{-\ell_{j-1}(x)}}\cdot\frac{1-q^{-a_j(x)}t^{\ell_j(x)+1}}{1-q^{-a_{j-1}(x)}t^{\ell_{j-1}(x)+1}}. \tag{67}$$

To prove (66), it suffices to show that

$$\frac{\Theta_{\sigma(j)}}{\Xi_{\sigma(j)}}=\frac{g_{\sigma(j)}}{g_{\sigma(j-1)}}\frac{1-qt}{(1-q)(1-t)},\quad\forall 1\le j\le n. \tag{68}$$

Without loss of generality, we argue for the case when $j = n$.

Suppose the shape of σ is given by the partition $(\lambda_1, \dots, \lambda_u)$ with transpose $(\lambda'_1, \dots, \lambda'_v)$. Here λ_j is the size of the j-th column and λ_i is the size of the i-th row. Assume the weight of the box labeled by n is $\chi_n = q^c t^r$. We have that:

$$\begin{aligned}
&\prod\nolimits_{x\in C_n} \frac{1-q^{1+a_n(x)}t^{-\ell_n(x)}}{1-q^{1+a_{n-1}(x)}t^{-\ell_{n-1}(x)}} \cdot \frac{1-q^{-a_n(x)}t^{\ell_n(x)+1}}{1-q^{-a_{n-1}(x)}t^{\ell_{n-1}(x)+1}} \\
=&\Omega\left(\sum_{x\in C_n}\left(q^{1+a_n(x)}(1-t)t^{-\ell_n(x)} + q^{-a_n(x)}(t-1)t^{\ell_n(x)}\right)\right) \\
=&\Omega\left(\sum_{j=1}^{r-1}\left(q^{\lambda_{j+1}-c}(1-t)t^{j-r} + q^{c+1-\lambda_{i+1}}(1-t^{-1})t^{r+1-j}\right)\right). \qquad (69)
\end{aligned}$$

Similarly,

$$\begin{aligned}
&\prod\nolimits_{x\in R_n} \frac{1-q^{1+a_n(x)}t^{-\ell_n(x)}}{1-q^{1+a_{n-1}(x)}t^{-\ell_{n-1}(x)}} \cdot \frac{1-q^{-a_n(x)}t^{\ell_n(x)+1}}{1-q^{-a_{n-1}(x)}t^{\ell_{n-1}(x)+1}} \\
=&\Omega\left(\sum_{i=1}^{c-1}\left(q^{c+1-i}(1-q^{-1})t^{r+1-\lambda'_{i+1}} + q^{i-c}(1-q)t^{\lambda'_{j+1}-r}\right)\right). \qquad (70)
\end{aligned}$$

By labeling on a Young diagram (cf. [BG13, Lemma 4.8.4]), we have that

$$A(q,t) := \sum_{i=0}^{c-1} q^i(1-q)t^{\lambda'_{i+1}} + \sum_{j=0}^{r-1} q^{\lambda_{j+1}}(1-t)t^j = -(1-q)(1-t)\sum_{i=1}^{n-1}\chi_i + 1 - q^c t^c =: B(q,t).$$

The equality

$$q^{-c}t^{-r}A(q,t) + q^{c+1}t^{r+1}A(q^{-1},t^{-1}) = q^{-c}t^{-r}B(q,t) + q^{c+1}t^{r+1}B(q^{-1},t^{-1})$$

expands to

$$q^{-c}t^{-r}\left(\sum_{i=0}^{c-1} q^i(1-q)t^{\lambda'_{i+1}} + \sum_{j=0}^{r-1} q^{\lambda_{j+1}}(1-t)t^j\right) + q^{c+1}t^{r+1}\left(\sum_{i=0}^{c-1} q^{-i}(1-q^{-1})t^{-\lambda'_{i+1}} + \sum_{j=0}^{r-1} q^{-\lambda_{j+1}}(1-t^{-1})t^{-j}\right) \qquad (71)$$

$$= -(1-q)(1-t)\sum_{i=1}^{n-1}\left(\frac{\chi_i}{\chi_n} + \frac{\chi_n}{\chi_i}\right) + q^{-c}t^{-r} - 1 + q^{c+1}t^{r+1} - qt. \qquad (72)$$

By (69) and (70), $\frac{g_{\sigma(n)}}{g_{\sigma(n-1)}} = (1-q)(1-t)\Omega\,((71))$. Moreover, $\frac{\Theta_{\sigma(n)}}{\Xi_{\sigma(n)}} = (1-qt)\Omega\,((72))$. Thus, (68) is proved. □

Remark 4.3. A similar formula appears in [KT22, Lemma 5.13].

5. Proofs of the main results

5.1. **Hodge filtrations and shift functors.** Equip eL_c and eL_{c+k} with Hodge filtrations from $\overline{\mathbf{N}}_c$.

Proposition 5.1. *When $c > 1$ the following isomorphism is filtered:*

$$\mathrm{eL}_{c+k} \cong {}_{c+k}P_c \otimes_{\mathrm{A}_c} \mathrm{eL}_c.$$

Proof. To prove the proposition, we switch to working with $X := \mathfrak{g} \times \mathbb{P}^{n-1}$. By Theorem 2.4, we have that

$$\mathrm{eL}_c \cong \Gamma\left(X, \mathbf{N}_c \boxtimes \mathcal{O}_{\mathbb{P}^{n-1}}(m)\right)^G.$$

Let $\mathcal{O}_X(nk)$ be the pullback of $\mathcal{O}_{\mathbb{P}^{n-1}}(nk)$ to X. Consider the sheaf of twisted differential operators $\mathcal{D}_{-c,X}$ on X associated to the line bundle $\mathcal{O}_{\mathbb{P}^{n-1}}(m)$. Define

$${}_{-c-k}\mathcal{D}_{-c,X} := \mathcal{O}_X(nk) \otimes_{\mathcal{O}_X} \mathcal{D}_{-c,X}.$$

It is shown in [GGS09, Lemma 6.7] that

$$\Gamma(X, {}_{-c-k}\mathcal{D}_{-c,X})^G \cong {}_{-c-k}Q_{-c}.$$

Write $\underline{\mathbf{N}}_c := \mathbf{N}_c \boxtimes \mathcal{O}_{\mathbb{P}^{n-1}}(m)$, which is a module over $\mathcal{D}_{-c,X}$. We have a commutative diagram:

$$\begin{array}{ccc} \Gamma\left({}_{-c-k}\mathcal{D}_{-c,X} \otimes_{\mathcal{D}_{-c,X}} \underline{\mathbf{N}}_c\right)^G & \xrightarrow{\sim} & \Gamma(X, \underline{\mathbf{N}}_{c+k})^G \\ \downarrow \sim & & \downarrow \\ \Omega_{c+k}\left({}_{c+k}P_c \otimes_{\mathrm{eH}_c\mathrm{e}} \mathrm{eL}_c\right) & \xrightarrow{\sim} & \Omega_{c+k}\left(\mathrm{eL}_{c+k}\right) \end{array}$$

By assumption, the right arrow is a filtered isomorphism. The left arrow is also a filtered isomorphism with respect to the tensor product filtrations. It remains to notice that the top arrow is also a filtered isomorphism:

$${}_{-c-k}\mathcal{D}_{-c,X} \otimes_{\mathcal{D}_{-c,X}} \underline{\mathbf{N}}_c \cong \mathcal{O}_X(nk) \otimes_{\mathcal{O}_X} \underline{\mathbf{N}}_c \cong \underline{\mathbf{N}}_{c+k}. \qquad \square$$

Therefore, by Proposition 2.7, we conclude that:

Corollary 5.2. *With respect to the gradings defined by $\widetilde{h}$, resp. h_c, and the Hodge filtration, the isomorphism $\Psi_c(\overline{\mathbf{N}}_c) \cong GS(\mathrm{eL}_c)$ holds in $\mathrm{Coh}^A(\mathrm{Hilb}^n)$.*

As a corollary, we recover the following result originally proved by Haiman [Hai98].

Corollary 5.3. *The punctual Hilbert scheme Hilb_0^n is Cohen-Macaulay.*

Proof. By [GS05, Proposition 1.7], $GS(\mathrm{eL}_{1/n}) = i_*\mathcal{O}_{\mathrm{Hilb}_0^n}$ where i is the embedding $\mathrm{Hilb}_0^n \hookrightarrow \mathrm{Hilb}^n$. Moreover, by [Sai88, Lemma 5.1.13], $\widetilde{\mathrm{gr}}(\overline{\mathbf{N}}_{1/n})$ is a Cohen-Macaulay module and hence so is the descent $\Psi_{1/n}(\overline{\mathbf{N}}_{1/n})$. Threfore, by Corollary 5.2, $GS(\mathrm{eL}_{1/n})$ is Cohen-Macaulay. $\square$

Proposition 5.1 also allows us to extend F from eL_c to L_c for all $c = \frac{m}{n} > 1$ by defining a tensor product filtration on L_c via (cf. [BEG03, Lemma 4.7])

$$\mathrm{L}_c \cong \mathrm{H}_c\delta\mathrm{e}_- \otimes_{\mathrm{A}_{c-1}} \mathrm{eL}_{c-1}. \tag{73}$$

In this way, the sub-filtration on eL_c coincides with the original F.

Note that we do not use the tensor product filtration coming from the Morita equivalence $\mathrm{L}_c \cong \mathrm{H}_c\mathrm{e} \otimes_{\mathrm{A}_c} \mathrm{eL}_c$.

5.2. **Bigraded characters.** Sort the formula (62) as $P_{m+n,n} \cdot 1 = \sum_{\lambda \vdash n} c^\lambda_{m,n} \widetilde{H}_\lambda$, with

$$c^\lambda_{m,n} = \left(\frac{(1-q)(1-t)}{(1-qt)}\right)^n \sum_{\sigma \in \mathrm{SYT}_\lambda} \frac{\prod_{i=1}^n \chi_{n-i+1}^{\lfloor im/n \rfloor - \lfloor (i-1)m/n \rfloor}}{g_\lambda \prod_{i=1}^{n-1}(1 - qt\frac{\chi_i}{\chi_{i+1}})} \Theta_\sigma.$$

We recall (46) which states that $(\widetilde{H}_\lambda, s_{(n)}) = 1$ for all $\lambda \vdash n$. On the other hand, $(\widetilde{H}_\lambda, s_{(1^n)}) = \prod_{i=1}^n \chi_i$ ([GH96, Theorem 2.2]). Therefore,

$$(P_{m+n,n} \cdot 1, s_{(n)}) = \sum_{\lambda \vdash n} c^\lambda_{m,n}, \tag{74}$$
$$(P_{m+n,n} \cdot 1, s_{(1^n)}) = \sum_{\lambda \vdash n} c^\lambda_{m+n,n}.$$

We will need the rational shuffle theorem. We will not define all the combinatorial notions involved but only state the relevant facts. Interested authors may refer to [GN15, 6.3] for detailed definitions.

Let DP be the set of (m,n)-Dyck paths and PF be the set of (m,n)-parking functions, which are labeled (m,n)-Dyck paths. In particular, there is a surjection from PF to DP.

Let $d = \frac{1}{2}(m-1)(n-1)$. Area and dinv, resp. $\overline{\text{dinv}}$, are two combinatorial statistics associated to each dyck path, resp. parking function, with integer values in the interval $[0,d]$ and z_π is a monomial in the variables z_i's. For a Dyck path D, when $\text{area}(D) = 0$, $\text{dinv}(D) = d$ and when $\text{dinv}(D) = 0$, $\text{area}(D) = d$.

Theorem 5.4. [GN15, Conjectures 6.1, 6.3], [Mel21, Theorem 5.8]

$$P_{m,n} \cdot 1 = \sum_{\pi \in PF} q^{\text{area}(\pi)} t^{\overline{\text{dinv}}(\pi)} z_\pi \tag{75}$$
$$\sum_{\lambda \vdash n} c^\lambda_{m,n} = \sum_{D \in DP} q^{\text{area}(D)} t^{\text{dinv}(D)} \tag{76}$$

Define a $\mathbb{C}(q,t)$-linear involution ι on K (cf. (43)) by sending s_λ to s_{λ^t}. For every $A \times S_n$-module M, one has that

$$\text{ch}_{A \times S_n}(\text{sgn} \otimes M) = \iota(\text{ch}_{A \times S_n}(M)).$$

The following theorem generalizes [BEG03, Conjecture 7.2, 7.3] proved in [GS06, Theorem 1.8].

Theorem 5.5. *Suppose $c = \frac{m}{n}$ for $m > 0$ coprime to n. With respect to the Hodge filtration and the Euler field h_c,*

(a) $\text{ch}_{A \times S_n}(\text{L}_{c+1}) = \text{ch}_{A \times S_n}\left(\text{sgn} \otimes \Gamma(\text{Hilb}^n, \mathcal{P} \otimes \Psi_c(\overline{\mathbf{N}}_c))\right) = \iota\left((P_{m+n,n} \cdot 1)(q, q^{-1}t)\right)$.

(b) $\text{ch}_A(\text{eL}_c) = \text{ch}_A \Gamma(\text{Hilb}^n, \Psi_c(\overline{\mathbf{N}}_c)) = \sum_{\lambda \vdash n} c^\lambda_{m,n}(q, q^{-1}t)$.

Proof. By localization formula (48),

$$\text{ch}_{A \times S_n} \Gamma(\text{Hilb}^n, \mathcal{P} \otimes \Psi_c(\overline{\mathbf{N}}_c)) = \sum_{\lambda \vdash n} \frac{\widetilde{H}_\lambda}{g_\lambda} \text{ch}_A(\Psi_c(\overline{\mathbf{N}}_c)|_{I_\lambda}).$$

Write $\pi : \widetilde{\text{Hilb}^n} \to \text{Hilb}^n$. Recall that

$$\pi^* \Psi_c(\overline{\mathbf{N}}_c) = (\mathscr{F}_c \boxtimes \mathcal{O}_V)|_{\widetilde{\text{Hilb}^n}},$$

and thus there exists integers a, b such that $\text{ch}_A(\Psi_c(\overline{\mathbf{N}}_c)|_{I_\lambda}) = q^a t^b \text{ch}_A(\mathscr{F}_c|_{\mathbf{e}_\lambda})$ for all $\lambda \vdash n$. Therefore, by Proposition 4.9,

$$\text{ch}_{A \times S_n}\left(\Gamma(\text{Hilb}^n, \mathcal{P} \otimes \Psi_c(\overline{\mathbf{N}}_c))\right) = q^a t^b P_{m,n} \cdot 1(q, q^{-1}t). \tag{77}$$

The change of variables $(q, q^{-1}t)$ comes from that h_c acts by weight $(1, -1)$.

Next, we show that with respect to the order filtration,

$$\text{gr}(\text{H}_c \text{e}_-) \cong \text{sgn} \otimes \Gamma(\mathfrak{C} \times V, \mathcal{R} \boxtimes \mathcal{O}_V)^{\overline{G}} \tag{78}$$

as $A \times S_n$-modules. Indeed, $\Gamma(\mathfrak{C} \times V, \mathcal{R} \boxtimes \mathcal{O}_V)^{\overline{G}} = \mathbb{C}[\mathfrak{IC} \times V]^{\overline{G}}$ and we claim that

$$\mathbb{C}[\mathfrak{IC} \times V]^{\overline{G}} \cong \mathbb{C}[\mathfrak{h} \times \mathfrak{h}].$$

By the definition of $\mathfrak{IC}$, it suffices to show that $\mathbb{C}[\mathfrak{C} \times V]^{\overline{G}} \cong \mathbb{C}[\mathfrak{h} \times \mathfrak{h}]^W$.

Consider the variety $\Lambda' := \{(X, Y, i, j) \in \mathfrak{g} \times \mathfrak{g} \times V \times V^* | [X, Y] + ij = 0\}$. By [GG06, Lemma 2.8.3], $\mathbb{C}[\Lambda']^{\overline{G}} \cong \mathbb{C}[\mathfrak{h} \times \mathfrak{h}]^W$. Therefore, we may consider the following natural maps:

$$\mathbb{C}[\Lambda']^{\overline{G}} \twoheadrightarrow \mathbb{C}[\mathfrak{C} \times V]^{\overline{G}} \twoheadrightarrow \mathbb{C}[\mathfrak{C}]^{\overline{G}} \cong \mathbb{C}[\mathfrak{h} \times \mathfrak{h}]^W.$$

Since the composition is an isomorphism, both surjections are isomorphisms too. Thus the claim holds. The identification (78) follows from the isomorphism $\mathrm{gr}(\mathrm{H}_c \mathrm{e}_-) \cong \mathrm{sgn} \otimes \mathbb{C}[\mathfrak{h} \times \mathfrak{h}]$.

We now consider the following commutative diagram:

$$\begin{array}{ccc}
\mathrm{sgn} \otimes \Gamma\left(\mathfrak{C} \times V, (\mathcal{R} \otimes \underline{\mathcal{F}_c}) \boxtimes \mathcal{O}_V\right)^{\overline{G}} & \xrightarrow{\sim} & \mathrm{sgn} \otimes \Gamma(\mathfrak{C} \times V, \mathcal{R} \boxtimes \mathcal{O}_V)^{\overline{G}} \otimes_{\mathbb{C}[\mathfrak{C} \times V]^{\overline{G}}} \Gamma(\mathfrak{C} \times V, \underline{\mathcal{F}_c} \boxtimes \mathcal{O}_V)^{\overline{G}} \\
\downarrow & & \downarrow \sim \\
\mathrm{sgn} \otimes \Gamma\left(\widetilde{\mathrm{Hilb}^n}, (\mathcal{R} \otimes \mathcal{F}_c) \boxtimes \mathcal{O}_V\right)^{\overline{G}} & & \mathrm{gr}(\mathrm{H}_{c+1} \mathrm{e}_-) \otimes_{\mathrm{gr}(\mathrm{A}_c)} \Gamma(\mathfrak{C} \times V, \widetilde{\mathrm{gr}}\ \overline{\mathbf{N}}_c)^{\overline{G}} \\
\downarrow \sim & & \downarrow \sim \\
\mathrm{sgn} \otimes \Gamma(\mathrm{Hilb}^n, \mathcal{P} \otimes \Psi_c(\overline{\mathbf{N}}_c)) & & \mathrm{gr}(\mathrm{H}_{c+1} \mathrm{e}_-) \otimes_{\mathrm{gr}(\mathrm{A}_c)} \mathrm{gr}(\mathrm{eL}_c) \\
\vdots & & \downarrow \\
\mathrm{gr}(\mathrm{L}_{c+1}) & \xleftarrow{\sim} & \mathrm{gr}(\mathrm{H}_{c+1} \mathrm{e}_- \otimes_{\mathrm{A}_c} \mathrm{eL}_c)
\end{array}$$

By (77),

$$\dim \Gamma(\mathrm{Hilb}^n, \mathcal{P} \otimes \Psi_c(\overline{\mathbf{N}}_c)) = (P_{m+n,n} \cdot 1, e_1^n)|_{q=t=1}$$

which equals the number of $(m+n, n)$-parking function by (75). On the other hand, $\dim(\mathrm{L}_{c+1}) = (m+n)^{n-1}$ ([BEG03, Theorem 1.6]), which is the number of $(m+n, n)$-parking functions. Therefore the injection and surjection in the diagram above have to be both isomorphisms. By reversing some of the arrows, we obtain the isomorphism indicated by the dotted arrow. Hence the first equality of (a) holds.

The first equality of (b) follows from Corollary 5.2. Moreover, by (73), $\mathrm{eL}_c = (\mathrm{L}_{c+1})^{\mathrm{sgn}}$. By (74),

$$(\iota(P_{m,n} \cdot 1), s_{(1^n)}) = (P_{m,n} \cdot 1, s_{(n)}) = \sum_{\lambda \vdash n} c_{m,n}^{\lambda}$$

and hence $\mathrm{ch}_A(\mathrm{eL}_c) = q^a t^b \sum_{\lambda \vdash n} c_{m,n}^{\lambda}(q, q^{-1}t)$.

It remains to show $q^a t^b = 1$. By [BEG03, Theorem 1.10], the highest, resp. lowest weight of eL_c under the action of h_c is d, resp. $-d$. The highest, resp. lowest, q-degree in $\sum_{\lambda \vdash n} c_{m,n}^{\lambda}(q, q^{-1}t)$ is d, resp. $-d$, as well. Therefore $q^a t^b = 1$. □

5.3. **Link homology.** The following theorem builds on [EH19, Hog17]:

Theorem 5.6. *([Mel22, Corollary 3.4]) Up to a constant factor, the triply graded Euler characteristic* $\mathrm{ch}_{a,q,t}(\mathrm{HHH}(T_{m,n}))$ *equals* $\sum a^{2i}(P_{m,n}, s_{(i,1^{n-i})})(qt, qt^{-1})$.

Definition 5.1. The Kazhdan filtration associated to the Hodge filtration and h_c is defined by

$$K_i \mathrm{L}_c = \sum_{2j+k \leq i} F_j \mathrm{L}_c(k)$$

As a corollary of Theorem 5.5 and Theorem 5.6, we conclude Theorem A.

Theorem 5.7. *For $m > n$ and $(m,n) = 1$, there is a triply graded isomorphism when $m > n$:*

$$\bigoplus_{i,j,k} \mathrm{HHH}^{i,j,k}(T_{m,n}) \cong \bigoplus_{i} \mathrm{Hom}_{S_n}\left(\wedge^i \mathfrak{h}, \bigoplus_{j,k} \mathrm{gr}_j^K\left(\mathrm{L}_{m/n}(k)\right)\right). \tag{79}$$

5.4. **Filtrations.**

5.4.1. *Inductive filtrations.* Since the left hand side of (79) is m,n-symmetric, we see that

Corollary 5.8. *Kazhdan filtrations K on $\mathrm{eL}_{m/n}$ and $\mathrm{eL}_{\frac{n}{m}}$ are compatible with the isomorphism $\mathrm{eL}_{m/n} \cong \mathrm{eL}_{n/m}$.*

Consider a partial order on the positive rational numbers in the following way: for coprime pairs (m,n), $\frac{m}{n} \prec \frac{m+n}{n}$; if $n < m$, then $\frac{m}{n} \prec \frac{n}{m}$.

On can always go from $c = \frac{m}{n}$ to $\frac{1}{n'}$ for some integer $n' > 1$ through a chain of rational numbers decreasing under the order $(\mathbb{Q}_{>0}, \prec)$.

Definition 5.2. [GORS14, Theorem 4.1] We define a filtration F^{ind} inductively as follows:

$$0 = F^{\mathrm{ind}}_{-1}\mathrm{eL}_{\frac{1}{n}} \subset F^{\mathrm{ind}}_{0}\mathrm{eL}_{\frac{1}{n}} = \mathrm{eL}_{\frac{1}{n}} = \mathrm{L}_{\frac{1}{n}}.$$

Next, F^{ind} is defined inductively on L_c under the order $(\mathbb{Q}_{>0}, \prec)$ using the isomorphisms:

$$\begin{aligned} &\text{when } m,n>1, && \mathrm{e}_n\mathrm{L}_{m/n} \cong \mathrm{e}_m\mathrm{L}_{\frac{n}{m}} && \text{[CEE09, 8.2]} \\ &\text{when } c>1, && \mathrm{L}_c \cong \mathrm{H}_c\delta\mathrm{e}_- \otimes_{A_{c-1}} \mathrm{eL}_{c-1} && \text{[BEG03, Lemma 4.7]} \end{aligned} \tag{80}$$

The filtration on the right hand side of (80) is the tensor product filtration. We endow H_c with the order filtration.

Let F^{H} denote the Hodge filtration. Combining Proposition 5.1 and Corollary 5.8, we obtain that

Proposition 5.9. *$F^{\mathrm{H}}_j\mathrm{L}_c = F^{\mathrm{ind}}_j\mathrm{L}_c$ when $c > 1$ and $F^{\mathrm{H}}_j\mathrm{eL}_c = F^{\mathrm{ind}}_j\mathrm{eL}_c$ when $c > 0$.*

5.4.2. *Algebraic filtrations.* On H_c we have the Fourier transform defined by

$$\Phi_c(x_i) = y_i, \quad \Phi_c(y_i) = -x_i, \quad \Phi_c(w) = w \tag{81}$$

which defines the Dunkl bilinear form

$$(-,-)_c : \mathbb{C}[\mathfrak{h}] \times \mathbb{C}[\mathfrak{h}] \to \mathbb{C}, \quad (f,g)_c = [\Phi_c(f)g]|_{x_i=0}.$$

When $c = \frac{m}{n} > 0$, with m,n coprime, $(-,-)_c$ has a nonzero kernel I_c and the resulting quotient $\mathbb{C}[\mathfrak{h}]/I_c$ is exactly isomorphic to L_c ([DO03, Proposition 2.34]). Inside $\mathbb{C}[\mathfrak{h}]$, take the ideal $\mathfrak{a} := (\mathbb{C}[\mathfrak{h}]^W_+)$. Let β_c be a nonzero highest weight vector in L_c and let $^{\perp_c}$ denote orthogonal complement with respect to $(-,-)_c$.

Definition 5.3.

(1) The $\mathfrak{a}$-filtration is defined by $F^{\mathfrak{a}}_i\mathrm{L}_c = \Phi_c[(\mathfrak{a}^{i+1})^{\perp_c}]\beta_c$.
(2) The algebraic filtration is the Kazhdan filtration associated to $F^{\mathfrak{a}}$ and h_c: $F^{\mathrm{alg}}_i(\mathrm{L}_c) = \sum_{2j+k\le i} F^{\mathfrak{a}}_j\mathrm{L}_c(k)$.

Proposition 5.10. *For $m > 0$ and $(m,n) = 1$, there is a triply graded isomorphism:*

$$\bigoplus_{i,j,k} \mathrm{HHH}^{i,j,k}(T_{m,n}) \cong \bigoplus_{i} \mathrm{Hom}_{S_n}\left(\wedge^i \mathfrak{h}, \bigoplus_{j,k} \mathrm{gr}_j^{\mathrm{alg}}\left(\mathrm{L}_{m/n}(k)\right)\right). \tag{82}$$

Proof. It is shown in [Ma24] that $F^{\mathrm{ind}} = F^{\mathfrak{a}}$. This plus Proposition 5.9 and Corollary 5.7 implies that the proposition holds when $m > n$. Denote the right hand side of (82) by $\mathfrak{H}_{m,n}$, it remains to show a triply graded isomorphism $\mathfrak{H}_{m,n} \cong \mathfrak{H}_{n,m}$.

By [Gor13, Corollary 1.1], for all $k \geq 0$ there is an isomorphism

$$\mathrm{Hom}_{S_n}(\wedge^k \mathbb{C}^{n-1}, \mathrm{L}_{m/n}) \cong \mathrm{Hom}_{S_m}(\wedge^k \mathbb{C}^{m-1}, \mathrm{L}_{\frac{n}{m}}) \tag{83}$$

via identifications with spaces of differential forms on a zero-dimensional moduli space associated with the plane curve singularity $x^m = y^n$. By [GORS14, Proposition 1.5], (83) is a bigraded isomorphism with respect to the algebraic filtration and h_c. □

Appendix A. Fourier transforms of cuspidal D-modules

We prove an auxilary result about the Fourier transform of the cuspidal character D-module. Although unrelated to the main results of the paper, this may be of independent interest. In particular, it implies the q,t-symmetry: $(P_{m,n}\cdot 1)(q,t)=(P_{m,n}\cdot 1)(t,q)$.

A.1. **Invariance under the Fourier transform.** The map $\mathfrak{g}\times\mathfrak{g}^*\to\mathfrak{g}^*\times\mathfrak{g}$ sending $(x,x^*)\mapsto(x^*,-x)$ induces an isomorphism $\sigma_1:\mathcal{D}(\mathfrak{g})\cong\mathcal{D}(\mathfrak{g}^*)$.

Further identitifying $\sigma_2:\mathcal{D}(\mathfrak{g}^*)\cong\mathcal{D}(\mathfrak{g})$ via a non-degenerate bilinear form $\mathfrak{g}\cong\mathfrak{g}^*$, we have obtain the Fourier transform induced by $\sigma:=\sigma_2\circ\sigma_1$:

$$\mathbb{F}:\mathcal{D}_{\mathfrak{g}}-\mathrm{mod}\to\mathcal{D}_{\mathfrak{g}}-\mathrm{mod}.$$

Proposition A.1. *As $\mathcal{D}_{\mathfrak{g}}$-modules, $\mathbb{F}\mathbf{N}_c\cong\mathbf{N}_c$.*

Proof. Put $\iota:T^*\mathfrak{g}\to T^*\mathfrak{g},(x,x^*)\mapsto(x^*,-x)$. Then

$$SS(\mathbb{F}(\mathbf{N}_c))=\iota(SS(\mathbf{N}_c))\subset\mathscr{N}\times\mathscr{N}.$$

Therefore, $\mathbb{F}(\mathbf{N}_c)$ is again cuspidal is determined by the monodromy of its restriction to $\mathscr{N}_r$. The proposition hence follows from the following:

Claim: There is an isomorphism $\iota^\dagger(\mathbb{F}(\mathbf{N}_c))\cong\mathcal{F}_c$ for $\iota:\mathscr{N}_r\hookrightarrow\mathfrak{g}$.

Recall that the cuspidal character D-module $\mathbf{N}_c$ can be expressed as $\widetilde{\mathrm{Ind}}_B^G\mathbf{L}_c$. Therefore $\mathbb{F}(\mathbf{N}_c)=\widetilde{\mathrm{Ind}}_B^G\mathbb{F}(\mathbf{L}_c)$.

Since

$$\Gamma(\mathfrak{g},\mathbf{L}_c)=\mathcal{D}(\mathfrak{g})/(\mathcal{D}(\mathfrak{g})\cdot O(\mathfrak{b}_-)+\sum_i\mathcal{D}(\mathfrak{g})(x_i\partial_{x_i}-ic)+\mathcal{D}(\mathfrak{g})\cdot S([\mathfrak{n},\mathfrak{n}])) \tag{84}$$

we see that

$$\Gamma(\mathfrak{g},\mathbb{F}(\mathbf{L}_c))=\mathcal{D}(\mathfrak{g})/(\mathcal{D}(\mathfrak{g})\cdot S(\mathfrak{b})+\sum_i\mathcal{D}(\mathfrak{g})(\partial_{y_i}y_i+ic)+\mathcal{D}(\mathfrak{g})\cdot O([\mathfrak{n}_-,\mathfrak{n}_-])). \tag{85}$$

Consider the standard $\mathfrak{sl}_2$-triple $\underline{e},\underline{f},\underline{h}$ (defined by (26)). Then (84) and (85) tells us that $\mathbf{L}_c$ is the minimal extension of an integrable connection supported on $B\cdot E$ and $\mathbb{F}(\mathbf{L}_c)$ is the minimal extension of an integrable connectionm supported on $B\cdot F$ defined by a horizontal section

$$y_1^{-c}y_2^{-2c}\cdots y_{n-1}^{-(n-1)c}. \tag{86}$$

Moreover, $\mathbf{L}_c|_{T\cdot E}$ is a local system on $T\cdot E$ such that

$$\mathcal{F}_c=\widetilde{\mathrm{Ind}}_T^G i^E_\dagger(i^E)^\dagger\iota^\dagger\mathbf{L}_c$$

with $i^E:T\cdot E\to\mathscr{N}_r$. Similarly, $\mathbb{F}(\mathbf{N}_c)|_{T\cdot F}$ is a local system on $T\cdot F$ such that

$$(\mathbb{F}(\mathbf{N}_c))|_{\mathscr{N}_r}=\widetilde{\mathrm{Ind}}_T^G i^F_\dagger(i^F)^\dagger\iota^\dagger\mathbb{F}(\mathbf{L}_c)$$

with $i^F:T\cdot F\to\mathscr{N}_r$.

It suffices to show that $(i^F)^\dagger\mathcal{F}_c=(i^F)^\dagger\iota^\dagger\mathbb{F}(\mathbf{L}_c)$. Recall that the pullback of $\mathcal{F}_c$ along the fibration $q:U\to\mathscr{N}_r$ is $\mathcal{E}_c$ and $\mathcal{E}_c$ is defined by the horizontal section s^c (eq. (7)).

$$s^c|_{T\cdot F}=v_1^{nc}y_1^{(n-1)c}y_2^{(n-2)c}\cdots y_{n-1}^c.$$

Therefore, $(i^F)^\dagger\mathcal{F}_c$ is defined by the horizontal section

$$y_1^{(n-1)c}y_2^{(n-2)c}\cdots y_{n-1}^c.$$

Finally, the lemma follows from the observation that the functions $y_1^{(n-1)c}y_2^{(n-2)c}\cdots y_{n-1}^{c}$ and $y_1^{-c-1}y_2^{-2c-1}\dots y_{n-1}^{-(n-1)c-1}$ (eq. (86)) define the same integrable connection on $T\cdot F$ as

$$((n-1)c,(n-2)c,\cdots,c)-(m,m,\cdots,m)=(-c,-2c,\cdots,-(n-1)c). \qquad \square$$

A.2. **An explicit description of the Fourier transform.** This subsection comes from an observation of V. Ginzburg.

Let $x=(x_{ij})$ be the standard coordinates of $\mathfrak{gl}_n$ and $(\partial)=(\partial_{x_{ij}})_{1\le i,j\le n}$. Take

$$e:=\frac{1}{2}\mathrm{tr}(x^2),\quad f:=-\frac{1}{2}\mathrm{tr}(\partial^2),\quad h=\sum_{1\le i,j\le n}x_{ij}\partial_{x_{ij}}+(n^2-1)/2. \tag{87}$$

Clearly, $[e,f]=h$, $[h,e]=2e$ and $[h,f]=-2f$. Notice that $[e,-],[f,-],[h,-]$ all preserve the homogeneous components of $\mathcal{D}(\mathfrak{g})$ (with $\deg(x_{ij})$=$\deg(\partial_{x_{ij}})$=1). In other words, this $\mathfrak{sl}_2$-action on $\mathcal{D}(\mathfrak{g})$ is locally finite and thus integrable. Moreover, $e-f$ acts on $\mathcal{D}(\mathfrak{g})$ via

$$[e-f,x_{ij}]=[-f,x_{ij}]=\partial_{x_{ij}},\quad [e-f,\partial_{x_{ij}}]=[e,\partial_{x_{ij}}]=-x_{ij}. \tag{88}$$

Hence its exponential $\mathrm{Ad}\, e^{\frac{i\pi}{2}(e-f)}$ gives exactly the Fourier transform σ.

On the other hand, any $\mathcal{D}(\mathfrak{g})$-module inherits such an $\mathfrak{sl}_2$-action. By [CEE09, Example 63], the action of $\{e,f,h\}$ on $\mathbf{N}_c$ is locally finite and thus also integrable. Denote the action of $e^{\frac{i\pi}{2}(e-f)}$ on $\mathbf{N}_c$ by Φ. Then (88) implies that

$$\Phi(x_{ij}a)=\partial_{x_{ij}}\Phi(a),\quad \Phi(\partial_{x_{ij}}a)=-x_{ij}\Phi(a),\quad \forall a\in\mathbf{N}_c, 1\le i,j\le n.$$

Therefore, Φ gives an explicit isomorphism between $\mathbf{N}_c$ and $\mathbb{F}(\mathbf{N}_c)$.

Appendix B. Examples

Here are some calculation results of $\sum_{\lambda\vdash n}\frac{\mathrm{ch}_{q,t}((\mathrm{R}p_*\mathcal{A}_c)|_{\mathbf{e}_\lambda})}{g_\lambda}$ and $\sum_{\lambda\vdash n}\frac{\mathrm{ch}_{q,t}((\mathrm{R}p'_*\mathcal{A}'_c)|_{\mathbf{e}_\lambda})}{g_\lambda}$. In view of [KT22], setting $q\to 1$ these statistics are also Shalika germs in the corresponding cases.

B.1. $n=2$.

B.1.1. *Catalan* $\mathcal{A}'_{k+\frac{1}{2}}$.

$$\overbrace{\frac{q^k}{1-\frac{t}{q}}}^{\boxed{1}\boxed{2}}+\overbrace{\frac{t^k}{1-\frac{q}{t}}}^{\begin{smallmatrix}\boxed{2}\\ \boxed{1}\end{smallmatrix}}=\sum_{i=0}^{k}q^it^{k-i}$$

B.1.2. *Cuspdial* $\mathcal{A}_{k+\frac{1}{2}}$.

$$\overbrace{\frac{q^{k+1}}{1-\frac{t}{q}}}^{\boxed{1}\boxed{2}}+\overbrace{\frac{t^{k+1}(1-q)(1-qt)}{(1-t)(1-t^2)(1-\frac{q}{t})}}^{\begin{smallmatrix}\boxed{2}\\ \boxed{1}\end{smallmatrix}}+\overbrace{\frac{t^k(1-qt^2)}{(1-t^2)(1-\frac{1}{t})}}^{\begin{smallmatrix}\boxed{1}\\ \boxed{2}\end{smallmatrix}}=q(\sum_{i=0}^{k}q^it^{k-i})$$

B.2. $n = 3$.

B.2.1. *Catalan* $\mathcal{A}'_{\frac{2}{3}}$.

$$\overbrace{\frac{\begin{array}{|c|c|}\hline 2 & \\ \hline 1 & 3 \\ \hline \end{array}}{\left(1-\frac{t}{q}\right)\left(1-\frac{q^2}{t}\right)}}^{q} + \overbrace{\frac{\begin{array}{|c|c|}\hline 3 & \\ \hline 1 & 2 \\ \hline \end{array}}{\left(1-\frac{q}{t}\right)\left(1-\frac{t^2}{q}\right)}}^{t} + \overbrace{\frac{\begin{array}{|c|c|c|}\hline 1 & 2 & 3 \\ \hline \end{array}}{\left(1-\frac{t}{q}\right)\left(1-\frac{t}{q^2}\right)}}^{q} + \overbrace{\frac{\begin{array}{|c|}\hline 3 \\ \hline 2 \\ \hline 1 \\ \hline \end{array}}{\left(1-\frac{q}{t}\right)\left(1-\frac{q}{t^2}\right)}}^{t}$$
$$= q + t$$

B.2.2. *Cuspidal* $\mathcal{A}_{\frac{2}{3}}$.

$$\overbrace{\frac{\begin{array}{|c|c|c|}\hline 1 & 2 & 3 \\ \hline \end{array}}{(1-\frac{t}{q})\left(1-\frac{t}{q^2}\right)}}^{q^3}$$
$$+\overbrace{\frac{\begin{array}{|c|c|}\hline 3 & \\ \hline 1 & 2 \\ \hline \end{array}}{q\left(1-q^2\right)t(1-qt)}}^{q\left(1-q^2\right)t(1-qt)}{(1-t)\left(1-\frac{q^2}{t}\right)\left(1-\frac{t}{q}\right)\left(1-\frac{t^2}{q}\right)} + \overbrace{\frac{\begin{array}{|c|c|}\hline 1 & \\ \hline 2 & 3 \\ \hline \end{array}}{qt(1-qt)^2}}^{qt(1-qt)^2}{(1-t)^2\left(1-\frac{q}{t}\right)\left(1-\frac{t^2}{q}\right)} + \overbrace{\frac{\begin{array}{|c|c|}\hline 2 & \\ \hline 1 & 3 \\ \hline \end{array}}{q\left(1-qt^2\right)}}^{q\left(1-qt^2\right)}{\left(1-\frac{1}{t}\right)(1-t)\left(1-\frac{t^2}{q}\right)}$$
$$+\overbrace{\frac{\begin{array}{|c|}\hline 3 \\ \hline 2 \\ \hline 1 \\ \hline \end{array}}{(1-q)^2t^3(1-qt)^2}}^{(1-q)^2t^3(1-qt)^2}{(1-t)^2\left(1-t^2\right)^2\left(1-\frac{q}{t^2}\right)\left(1-\frac{q}{t}\right)} + \overbrace{\frac{\begin{array}{|c|}\hline 3 \\ \hline 1 \\ \hline 2 \\ \hline \end{array}}{(1-q)t^3(1-qt)\left(1-qt^2\right)}}^{(1-q)t^3(1-qt)\left(1-qt^2\right)}{\left(1-\frac{1}{t}\right)(1-t)\left(1-t^2\right)\left(1-t^3\right)\left(1-\frac{q}{t^2}\right)}$$
$$+\overbrace{\frac{\begin{array}{|c|}\hline 2 \\ \hline 3 \\ \hline 1 \\ \hline \end{array}}{(1-q)t^2(1-qt)\left(1-qt^2\right)}}^{(1-q)t^2(1-qt)\left(1-qt^2\right)}{\left(1-\frac{1}{t}\right)(1-t)\left(1-t^2\right)\left(1-t^3\right)\left(1-\frac{q}{t^2}\right)} + \overbrace{\frac{\begin{array}{|c|}\hline 1 \\ \hline 2 \\ \hline 3 \\ \hline \end{array}}{t\left(1-qt^2\right)\left(1-qt^3\right)}}^{t\left(1-qt^2\right)\left(1-qt^3\right)}{\left(1-\frac{1}{t^2}\right)\left(1-\frac{1}{t}\right)\left(1-t^2\right)\left(1-t^3\right)}$$
$$=q^2(q+t)$$

B.3. $n = 4$.

B.3.1. *Catalan* $\mathcal{A}'_{\frac{3}{4}}$.

$$\overbrace{\frac{\begin{array}{|c|c|c|}\hline 3 & & \\ \hline 1 & 2 & 4 \\ \hline \end{array}}{qt(1-t)\left(1-\frac{q^2}{t}\right)\left(1-\frac{t}{q}\right)^2\left(1-\frac{t^2}{q^2}\right)}}^{} + \overbrace{\frac{\begin{array}{|c|c|c|}\hline 2 & & \\ \hline 1 & 3 & 4 \\ \hline \end{array}}{qt\left(1-\frac{q}{t}\right)\left(1-\frac{t}{q}\right)\left(1-\frac{t^2}{q^2}\right)}}^{}$$

$$+\overbrace{\frac{\begin{array}{|c|c|}\hline 3 & 4 \\ \hline 1 & 2 \\ \hline \end{array}}{(1-q)qt\left(1-\frac{q}{t}\right)\left(1-\frac{q^2}{t}\right)\left(1-\frac{t}{q}\right)^2}}^{} + \overbrace{\frac{\begin{array}{|c|c|c|}\hline 4 & & \\ \hline 1 & 2 & 3 \\ \hline \end{array}}{q^3\left(1-\frac{t}{q^3}\right)\left(1-\frac{t}{q^2}\right)\left(1-\frac{t}{q}\right)}}^{} + \overbrace{\frac{\begin{array}{|c|c|c|c|}\hline 1 & 2 & 3 & 4 \\ \hline \end{array}}{q^3\left(1-\frac{q^3}{t}\right)\left(1-\frac{t}{q^2}\right)\left(1-\frac{t}{q}\right)}}^{}$$

$$+\overbrace{\frac{\begin{array}{|c|c|}\hline 4 & \\ \hline 2 & \\ \hline 1 & 3 \\ \hline \end{array}}{qt(1-t)\left(1-\frac{q^2}{t^2}\right)\left(1-\frac{q}{t}\right)^2\left(1-\frac{t^2}{q}\right)}}^{} + \overbrace{\frac{\begin{array}{|c|c|}\hline 4 & \\ \hline 3 & \\ \hline 1 & 2 \\ \hline \end{array}}{qt\left(1-\frac{q^2}{t^2}\right)\left(1-\frac{q}{t}\right)\left(1-\frac{t}{q}\right)}}^{}$$

$$+\overbrace{\frac{\begin{array}{|c|c|}\hline 2 & 4 \\ \hline 1 & 3 \\ \hline \end{array}}{qt(1-t)\left(1-\frac{q}{t}\right)^2\left(1-\frac{t}{q}\right)\left(1-\frac{t^2}{q}\right)}}^{} + \overbrace{\frac{\begin{array}{|c|c|}\hline 3 & \\ \hline 2 & \\ \hline 1 & 4 \\ \hline \end{array}}{t^3\left(1-\frac{q}{t^2}\right)\left(1-\frac{q}{t}\right)\left(1-\frac{q}{t^3}\right)}}^{} + \overbrace{\frac{\begin{array}{|c|}\hline 4 \\ \hline 3 \\ \hline 2 \\ \hline 1 \\ \hline \end{array}}{t^3\left(1-\frac{q}{t^3}\right)\left(1-\frac{q}{t^2}\right)\left(1-\frac{q}{t}\right)}}^{}$$

$$= q^3 + q^2t + qt + qt^2 + t^3$$

References

[Ach] Pramod Achar, *Equivariant mixed Hodge modules*, Lecture notes.

[Bal23] Ana Balibanu, *The partial compactification of the universal centralizer*, Selecta Math. (N.S.) **29** (2023), no. 5, Paper No. 85, 36.

[BE09] Roman Bezrukavnikov and Pavel Etingof, *Parabolic induction and restriction functors for rational Cherednik algebras*, Selecta Math. (N.S.) **14** (2009), no. 3-4, 397–425.

[BEG03] Yuri Berest, Pavel Etingof, and Victor Ginzburg, *Finite-dimensional representations of rational Cherednik algebras*, Int. Math. Res. Not. (2003), no. 19, 1053–1088.

[BFG06] Roman Bezrukavnikov, Michael Finkelberg, and Victor Ginzburg, *Cherednik algebras and Hilbert schemes in characteristic p*, Represent. Theory **10** (2006), 254–298.

[BG98] F Bergeron and G Garsia, *Science fiction and Macdonald's polynomials*, arXiv preprint math/9809128 (1998).

[BG13] Gwyn Bellamy and Victor Ginzburg, *Some combinatorial identities related to commuting varieties and Hilbert schemes*, Math. Ann. **355** (2013), no. 3, 801–847.

[BG15] ———, *Hamiltonian reduction and nearby cycles for mirabolic $\mathscr{D}$-modules*, Adv. Math. **269** (2015), 71–161.

[BL94] Joseph Bernstein and Valery Lunts, *Equivariant sheaves and functors*, Lecture Notes in Mathematics, vol. 1578, Springer-Verlag, Berlin, 1994.

[BO] Roman Bezrukavnikov and Andrei Okounkov, *Monodromy and derived equivalences*, unpublished.

[BS12] Igor Burban and Olivier Schiffmann, *On the Hall algebra of an elliptic curve, I*, Duke Math. J. **161** (2012), no. 7, 1171–1231.

[CEE09] Damien Calaque, Benjamin Enriquez, and Pavel Etingof, *Universal KZB equations: the elliptic case*, Algebra, arithmetic, and geometry: in honor of Yu. I. Manin. Vol. I, Progr. Math., vol. 269, Birkhäuser Boston, Boston, MA, 2009, pp. 165–266.

[CG10] Neil Chriss and Victor Ginzburg, *Representation theory and complex geometry*, Modern Birkhäuser Classics, Birkhäuser Boston, Ltd., Boston, MA, 2010.

[Che98] Jan Cheah, *Cellular decompositions for nested Hilbert schemes of points*, Pacific J. Math. **183** (1998), no. 1, 39–90.

[Che13] Ivan Cherednik, *Jones polynomials of torus knots via DAHA*, Int. Math. Res. Not. IMRN (2013), no. 23, 5366–5425.

[Del70] Pierre Deligne, *Équations différentielles à points singuliers réguliers*, Lecture Notes in Mathematics, vol. Vol. 163, Springer-Verlag, Berlin-New York, 1970.

[DO03] C. F. Dunkl and E. M. Opdam, *Dunkl operators for complex reflection groups*, Proc. London Math. Soc. (3) **86** (2003), no. 1, 70–108.

[EG02] Pavel Etingof and Victor Ginzburg, *Symplectic reflection algebras, Calogero-Moser space, and deformed Harish-Chandra homomorphism*, Invent. Math. **147** (2002), no. 2, 243–348.

[EGL15] Pavel Etingof, Eugene Gorsky, and Ivan Losev, *Representations of rational Cherednik algebras with minimal support and torus knots*, Adv. Math. **277** (2015), 124–180.

[EH19] Ben Elias and Matthew Hogancamp, *On the computation of torus link homology*, Compos. Math. **155** (2019), no. 1, 164–205.

[EKLS21] Pavel Etingof, Vasily Krylov, Ivan Losev, and José Simental, *Representations with minimal support for quantized Gieseker varieties*, Math. Z. **298** (2021), no. 3-4, 1593–1621.

[ESm88] Geir Ellingsrud and Stein Arild Strø mme, *On a cell decomposition of the Hilbert scheme of points in the plane*, Invent. Math. **91** (1988), no. 2, 365–370.

[FG10] Michael Finkelberg and Victor Ginzburg, *On mirabolic D-modules*, Int. Math. Res. Not. IMRN (2010), no. 15, 2947–2986.

[FT11] B. L. Feigin and A. I. Tsymbaliuk, *Equivariant K-theory of Hilbert schemes via shuffle algebra*, Kyoto J. Math. **51** (2011), no. 4, 831–854.

[GG06] Wee Liang Gan and Victor Ginzburg, *Almost-commuting variety, $\mathscr{D}$-modules, and Cherednik algebras*, IMRP Int. Math. Res. Pap. (2006), 26439, 1–54.

[GGOR03] Victor Ginzburg, Nicolas Guay, Eric Opdam, and Raphaël Rouquier, *On the category $\mathscr{O}$ for rational Cherednik algebras*, Invent. Math. **154** (2003), no. 3, 617–651.

[GGS09] V. Ginzburg, I. Gordon, and J. T. Stafford, *Differential operators and Cherednik algebras*, Selecta Math. (N.S.) **14** (2009), no. 3-4, 629–666.

[GH96] A. M. Garsia and M. Haiman, *A remarkable q,t-Catalan sequence and q-Lagrange inversion*, J. Algebraic Combin. **5** (1996), no. 3, 191–244.

[GH98] ———, *A random q, t-hook walk and a sum of Pieri coefficients*, J. Combin. Theory Ser. A **82** (1998), no. 1, 74–111.

[Gin00] Victor Ginzburg, *Principal nilpotent pairs in a semisimple Lie algebra. I*, Invent. Math. **140** (2000), no. 3, 511–561.

[Gin12] ———, *Isospectral commuting variety, the Harish-Chandra D-module, and principal nilpotent pairs*, Duke Math. J. **161** (2012), no. 11, 2023–2111.

[GN15] Eugene Gorsky and Andrei Neguţ, *Refined knot invariants and Hilbert schemes*, J. Math. Pures Appl. (9) **104** (2015), no. 3, 403–435.

[GN17] ———, *Infinitesimal change of stable basis*, Selecta Math. (N.S.) **23** (2017), no. 3, 1909–1930.

[GN24] ———, *Hecke categories, idempotents, and commuting stacks*, arXiv: 2406.05215.

[Gor12] Iain Gordon, *Macdonald positivity via the Harish-Chandra D-module*, Invent. Math. **187** (2012), no. 3, 637–643.

[Gor13] Evgeny Gorsky, *Arc spaces and DAHA representations*, Selecta Math. (N.S.) **19** (2013), no. 1, 125–140.

[GORS14] Eugene Gorsky, Alexei Oblomkov, Jacob Rasmussen, and Vivek Shende, *Torus knots and the rational DAHA*, Duke Math. J. **163** (2014), no. 14, 2709–2794.

[GS05] I. Gordon and J. T. Stafford, *Rational Cherednik algebras and Hilbert schemes*, Adv. Math. **198** (2005), no. 1, 222–274.

[GS06] ———, *Rational Cherednik algebras and Hilbert schemes. II. Representations and sheaves*, Duke Math. J. **132** (2006), no. 1, 73–135.

[GSV23] Nicolle González, José Simental, and Monica Vazirani, *Higher rank (q,t)-catalan polynomials, affine Springer fibers, and a finite rational shuffle theorem*, arXiv:2303.15694 (2023).

[Hai98] Mark Haiman, *t,q-Catalan numbers and the Hilbert scheme*, Discrete Math. **193** (1998), no. 1-3, 201–224.

[Hai01] ———, *Hilbert schemes, polygraphs and the Macdonald positivity conjecture*, J. Amer. Math. Soc. **14** (2001), no. 4, 941–1006.

[Hai02] ———, *Vanishing theorems and character formulas for the Hilbert scheme of points in the plane*, Invent. Math. **149** (2002), no. 2, 371–407.

[Hai03] ———, *Combinatorics, symmetric functions, and Hilbert schemes*, Current developments in mathematics, 2002, Int. Press, Somerville, MA, 2003, pp. 39–111.

[Hog17] Matthew Hogancamp, *Khovanov-rozansky homology and higher Catalan sequences*, arXiv:1704.01562 (2017).

[Hog18] ———, *Categorified Young symmetrizers and stable homology of torus links*, Geom. Topol. **22** (2018), no. 5, 2943–3002.

[HTT08] Ryoshi Hotta, Kiyoshi Takeuchi, and Toshiyuki Tanisaki, *D-modules, perverse sheaves, and representation theory*, japanese ed., Progress in Mathematics, vol. 236, Birkhäuser Boston, Inc., Boston, MA, 2008.

[Jon87] V. F. R. Jones, *Hecke algebra representations of braid groups and link polynomials*, Ann. of Math. (2) **126** (1987), no. 2, 335–388.

[Kas03] Masaki Kashiwara, *D-modules and microlocal calculus*, Translations of Mathematical Monographs, vol. 217, American Mathematical Society, Providence, RI, 2003.

[Kho07] Mikhail Khovanov, *Triply-graded link homology and Hochschild homology of Soergel bimodules*, Internat. J. Math. **18** (2007), no. 8, 869–885.

[KT22] Oscar Kivinen and Cheng-Chiang Tsai, *Shalika germs for tamely ramified elements in GL_n*, arXiv preprint arXiv:2209.02509 (2022).

[Lau83] Gérard Laumon, *Sur la catégorie dérivées des d-modules filtrés*, Algebraic geometry, Springer, 1983, pp. 151–237.

[Lus86] George Lusztig, *Character sheaves. IV*, Adv. in Math. **59** (1986), no. 1, 1–63.

[Lus87] ———, *Fourier transforms on a semisimple Lie algebra over $\mathbf{F}_q$*, Algebraic groups Utrecht 1986, Lecture Notes in Math., vol. 1271, Springer, Berlin, 1987, pp. 177–188.

[Ma24] Xinchun Ma, *Rational cherednik algebras and torus knot invariants*, arXiv:2402.18770 (2024).

[Mac95] I. G. Macdonald, *Symmetric functions and Hall polynomials*, second ed., Oxford Mathematical Monographs, The Clarendon Press, Oxford University Press, New York, 1995, With contributions by A. Zelevinsky, Oxford Science Publications.

[Mel21] Anton Mellit, *Toric braids and (m,n)-parking functions*, Duke Math. J. **170** (2021), no. 18, 4123–4169.

[Mel22] ———, *Homology of torus knots*, Geom. Topol. **26** (2022), no. 1, 47–70.

[Nak99] Hiraku Nakajima, *Lectures on Hilbert schemes of points on surfaces*, University Lecture Series, vol. 18, American Mathematical Society, Providence, RI, 1999.

[Neg14] Andrei Neguţ, *The shuffle algebra revisited*, Int. Math. Res. Not. IMRN (2014), no. 22, 6242–6275.

[Neg15a] ———, *Moduli of flags of sheaves and their K-theory*, Algebr. Geom. **2** (2015), no. 1, 19–43.

[Neg15b] ———, *Quantum Algebras and Cyclic Quiver Varieties*, ProQuest LLC, Ann Arbor, MI, 2015, Thesis (Ph.D.)–Columbia University.

[Neg22] ———, *W-algebras associated to surfaces*, Proc. Lond. Math. Soc. (3) **124** (2022), no. 5, 601–679.

[OY16] Alexei Oblomkov and Zhiwei Yun, *Geometric representations of graded and rational Cherednik algebras*, Adv. Math. **292** (2016), 601–706.

[Pop18] Mihnea Popa, *$\mathcal{D}$-modules in birational geometry*, Proceedings of the International Congress of Mathematicians—Rio de Janeiro 2018. Vol. II. Invited lectures, World Sci. Publ., Hackensack, NJ, 2018, pp. 781–806.

[Sai88] Morihiko Saito, *Modules de Hodge polarisables*, Publ. Res. Inst. Math. Sci. **24** (1988), no. 6, 849–995.

[Sai90] ———, *Mixed Hodge modules*, Publ. Res. Inst. Math. Sci. **26** (1990), no. 2, 221–333.

[Sai22] Takahiro Saito, *A description of monodromic mixed Hodge modules*, J. Reine Angew. Math. **786** (2022), 107–153.

[Sch] Christian Schnell, *An overview of Morihiko Saito's theory of mixed Hodge modules*, Representation theory, automorphic forms & complex geometry, Int. Press, Somerville, MA, pp. 27–80.

[SV13] Olivier Schiffmann and Eric Vasserot, *The elliptic Hall algebra and the K-theory of the Hilbert scheme of $\mathbb{A}^2$*, Duke Math. J. **162** (2013), no. 2, 279–366.

Email address: xinchma@gmail.com